材料科学的艺术：超材料



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主要内容

- 超材料的起源
- 超材料简介
- 经典超材料
- 超材料发展方向
- 材料科学的艺术—超材料

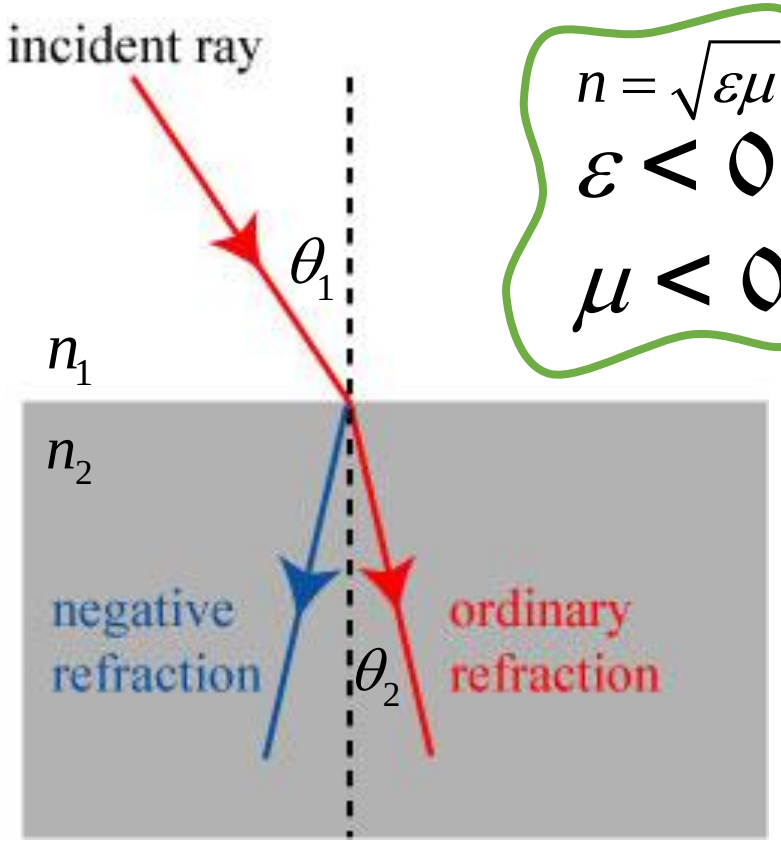
超材料的起源

- 负折射 (1968)

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$



V. Veselago



$$n = \sqrt{\varepsilon\mu}$$
$$\varepsilon < 0$$
$$\mu < 0$$



金属



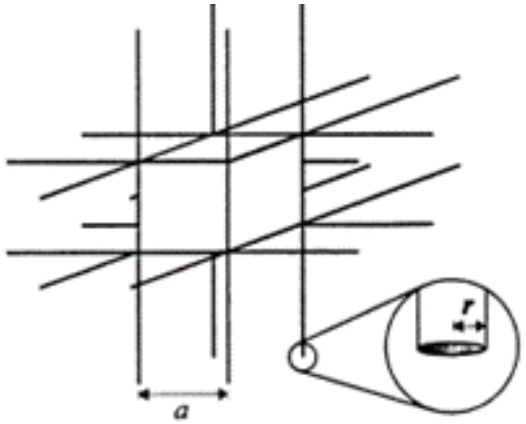
铁磁材料

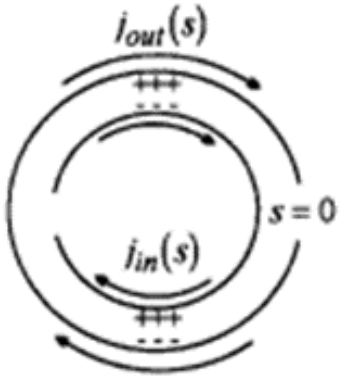
负介电常数与负磁导率

- 通过人工结构获得超常电磁参数 (1998-1999)

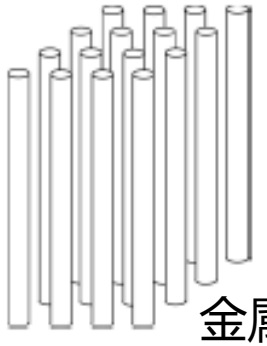


John. Pendry

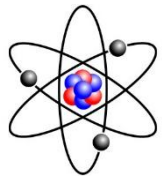




$$\varepsilon < 0$$
$$\mu < 0$$



金属线



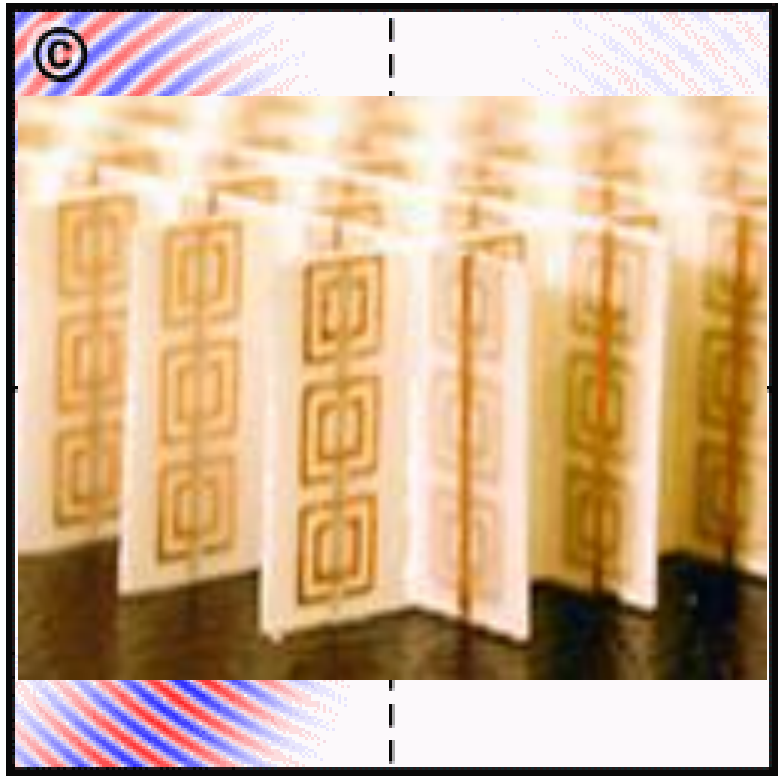


开口环

超材料的诞生

- 负折射率材料 (2001)





J. Pendry and D. R. Smith

什么是超材料

Meta is from the Greek word μετά, meaning "beyond" or "after", and the Latin word material, meaning "matter“ or "material"

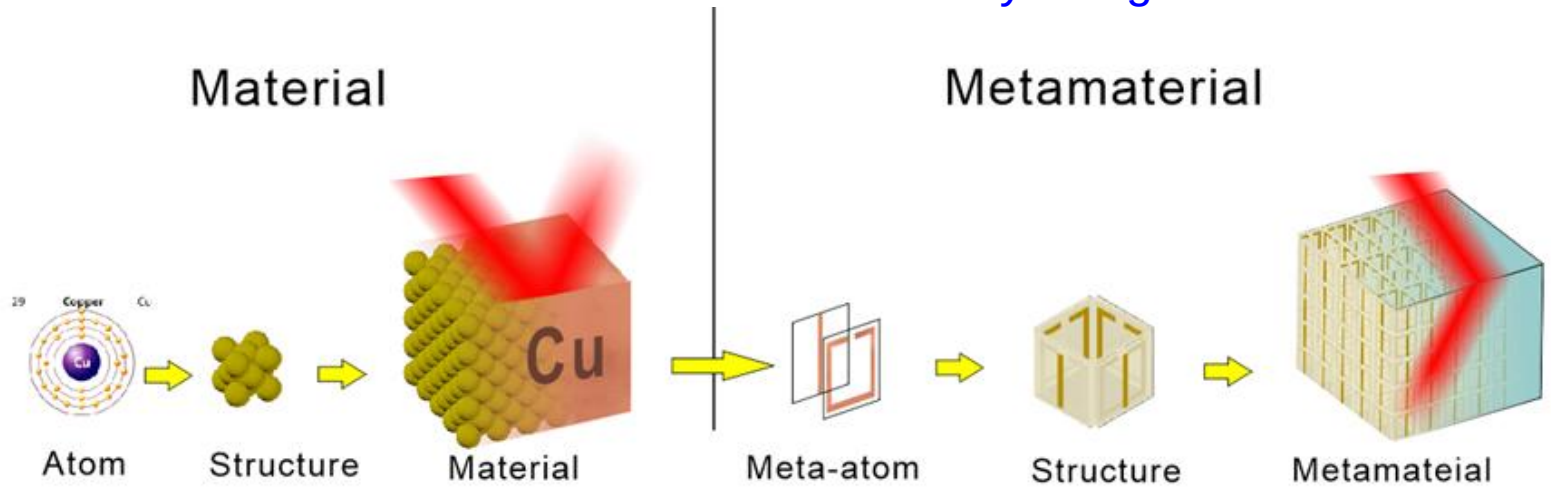
- 超·材料

通过人工设计的结构基元获得自然材料所不具有的超常属性的人工材料。

超材料的超常属性源于人工设计的结构，而并非构成结构的材料的本征属性。

Meta·materials

A **metamaterial** is a type of materials engineered to have a property, typically rarely observed in naturally occurring materials, that is derived not from the properties of the base materials but from their newly designed structures.

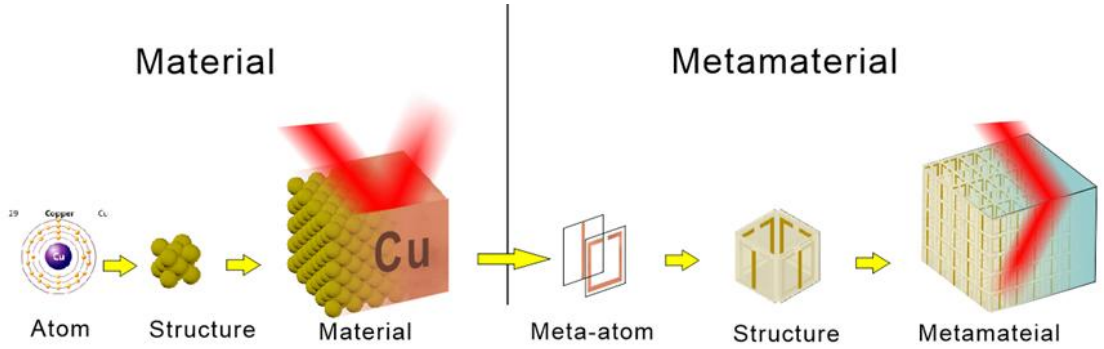


超材料与艺术

- 超材料

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艺术

艺术定义为“指凭借技巧、意愿、想象力、经验等综合人为因素的融合与平衡，以创作隐含美学的器物、环境、影像、动作或声音的表达模式，包括绘画、雕刻等具体化的创造，也包括文字、音乐、舞蹈、戏剧文学作品。

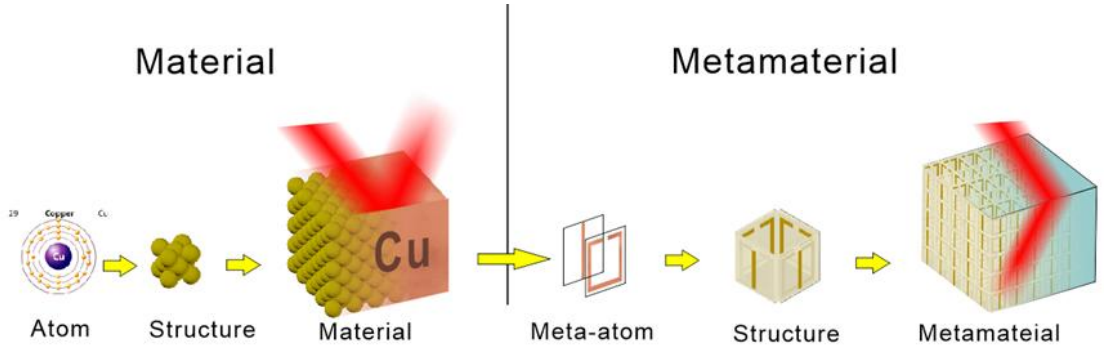
艺术来源于生活，高于生活。

超材料与艺术

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艺术来源于生活，高于生活。

超材料是一种新的研究范式

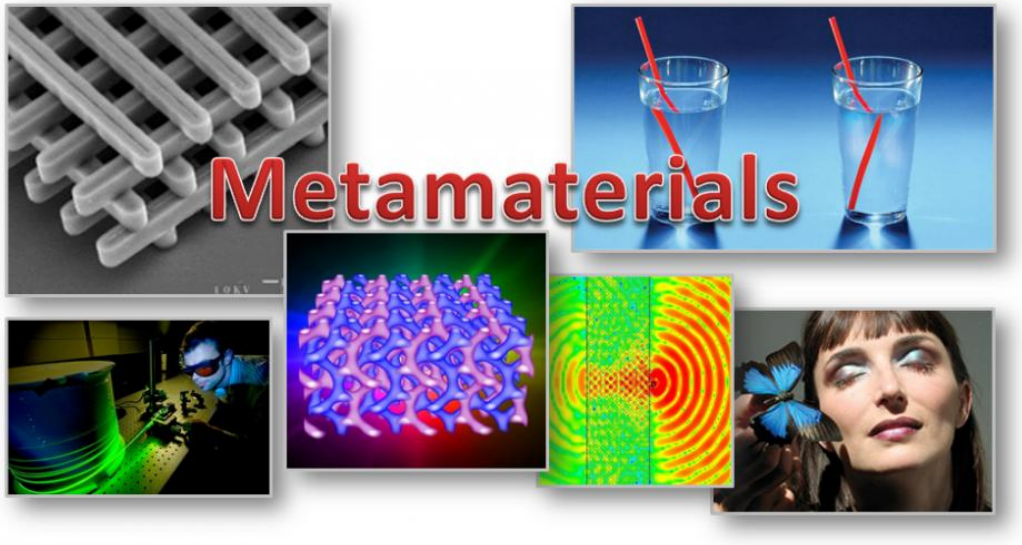
- 传统技术框架

面对需求找材料，一直面临有没有与好不好的问题。



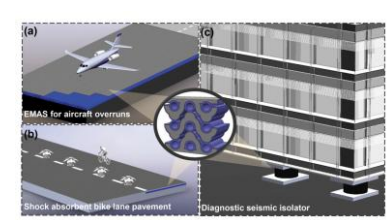
- 超材料

面对需求造材料，自然界没有的，靠人的设计获取。



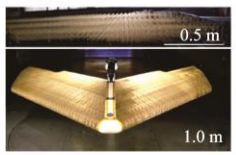
超材料的发展

- 超材料的多样性

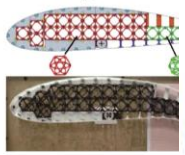






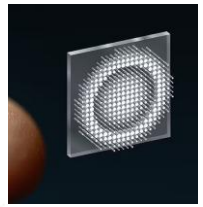


(c) Mechanical metamaterial wing

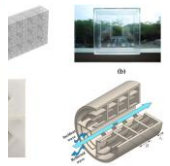


(d) Shape-shifting

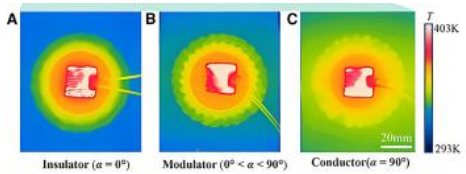


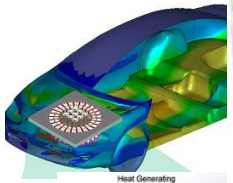


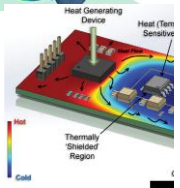
















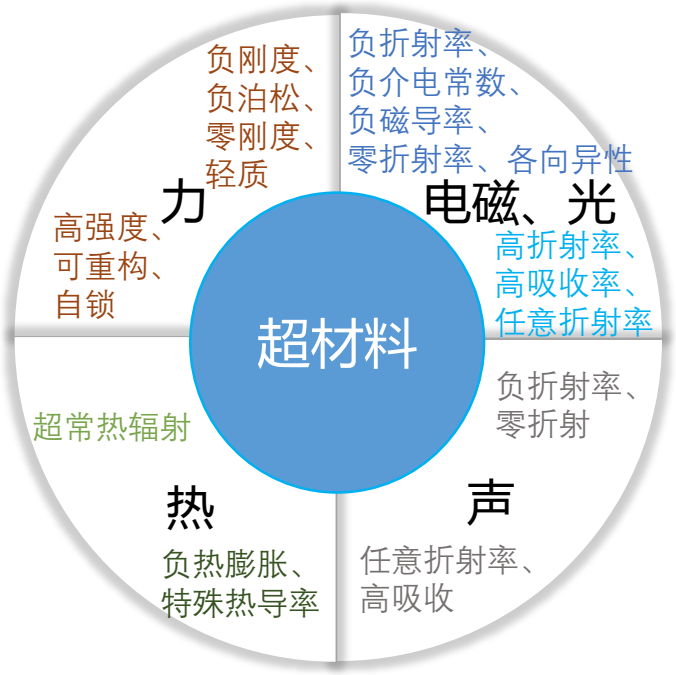
电磁波

物理学领域

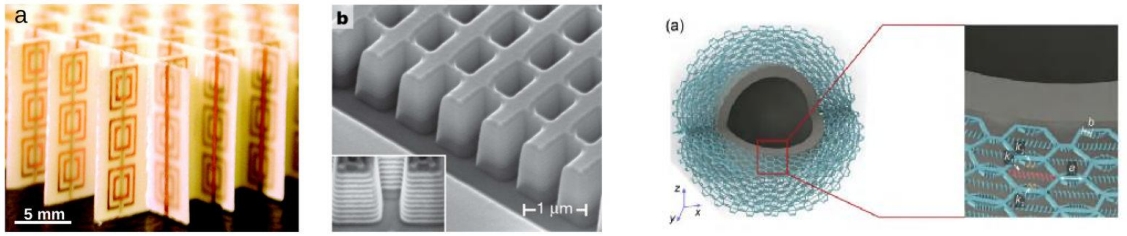
各行各业的技领域：信息技术、国防科技、土木工程、航空航天、生命健康

超材料的发展

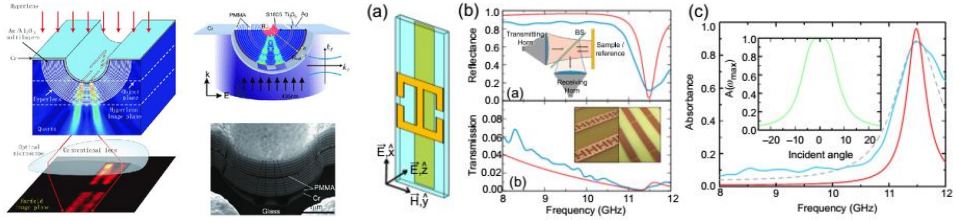
- 超材料的关键在于如何实现“超”



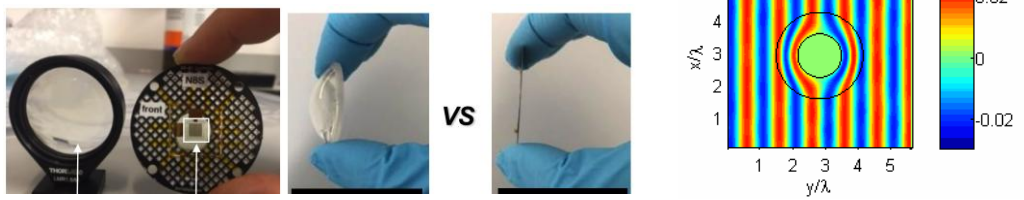
- 无中生有：　　如：负折射率、零刚度...



- 无与论比：　　如：强各向异性、强吸收...

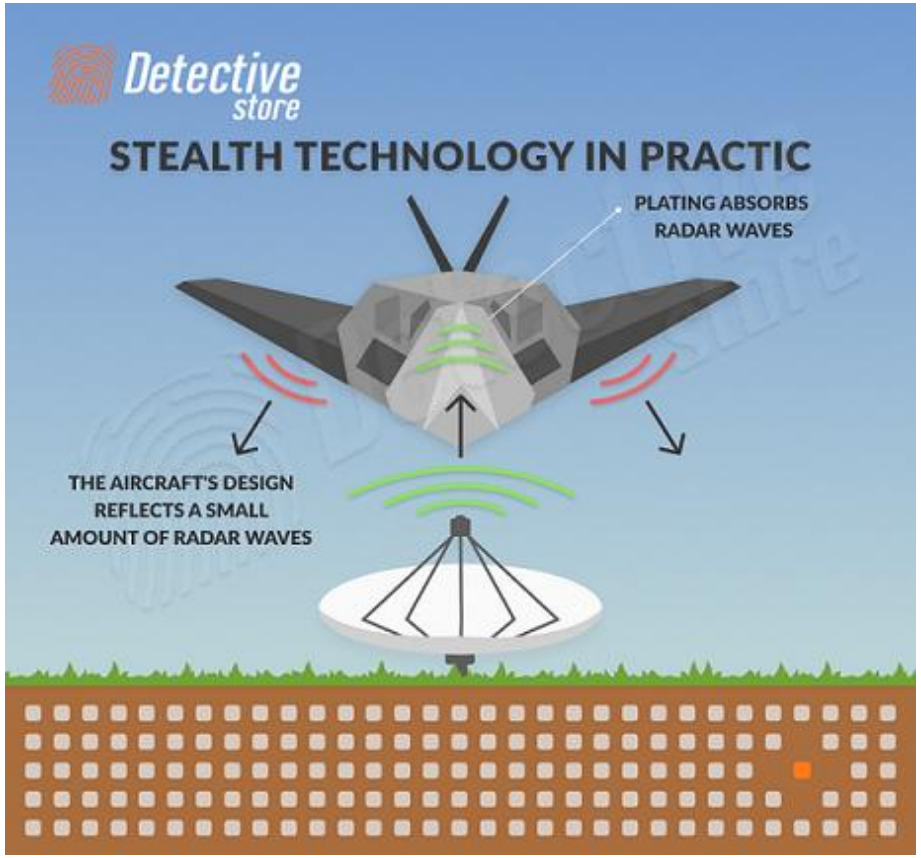


- 无所不能　　如：光学器件平面化、隐身



经典超材料

- 举个例子，需求: 隐身

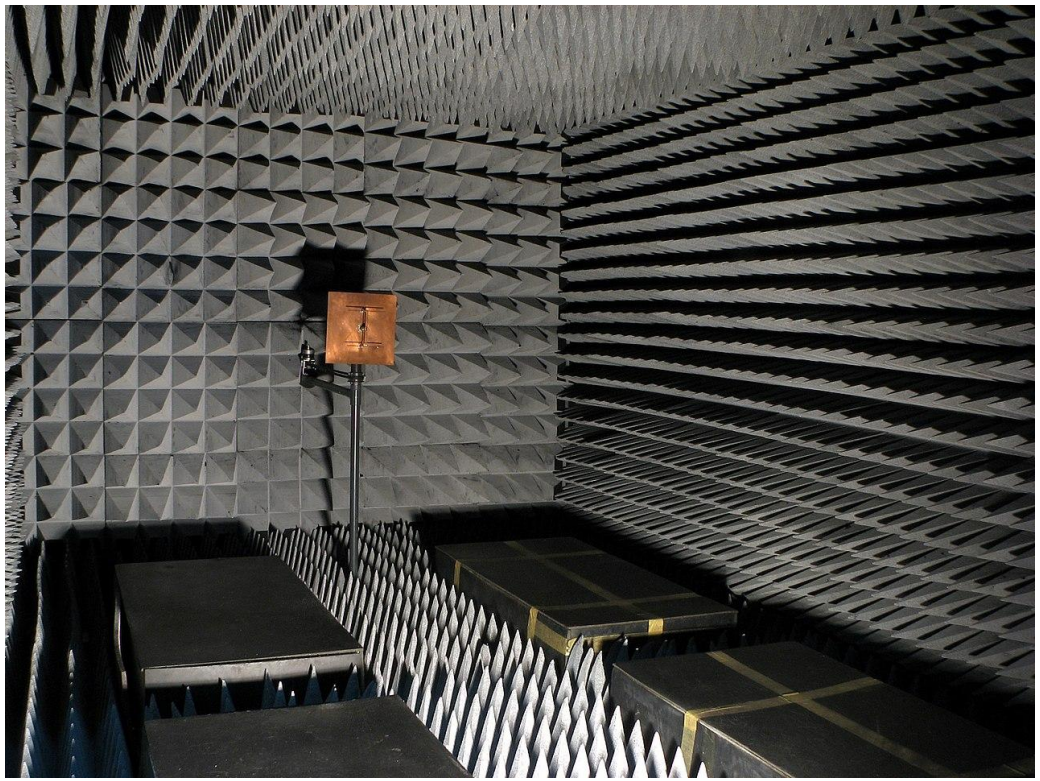




吸波材料

- 常规吸波材料

  吸波泡沫制成的角锥

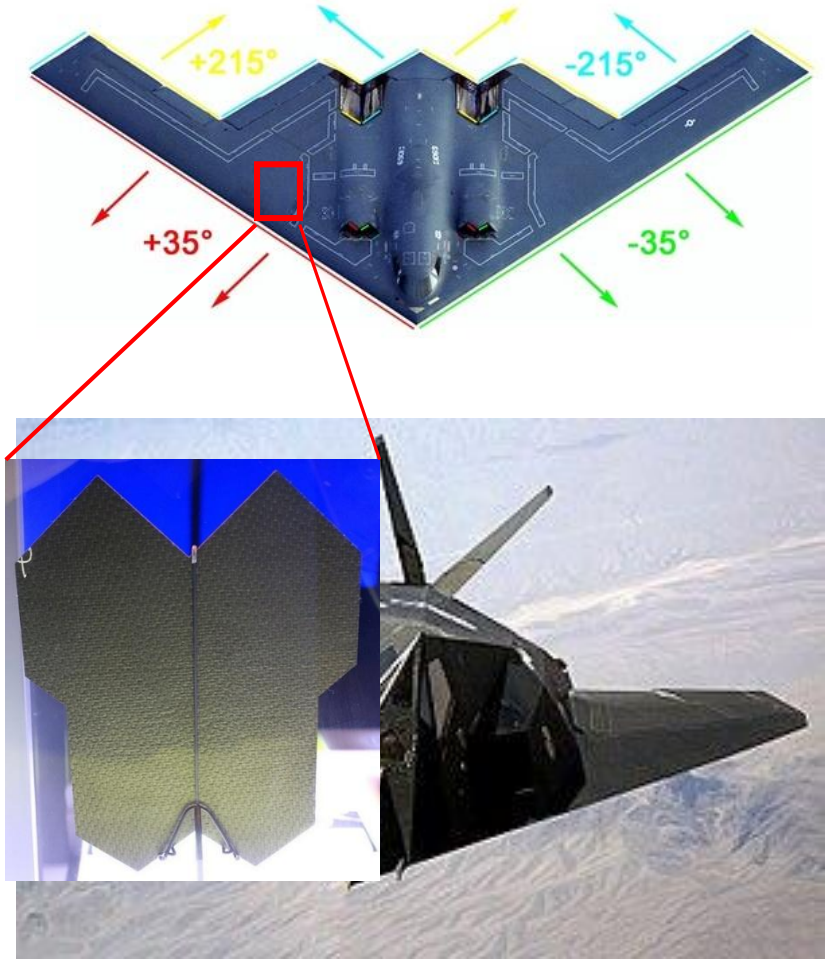


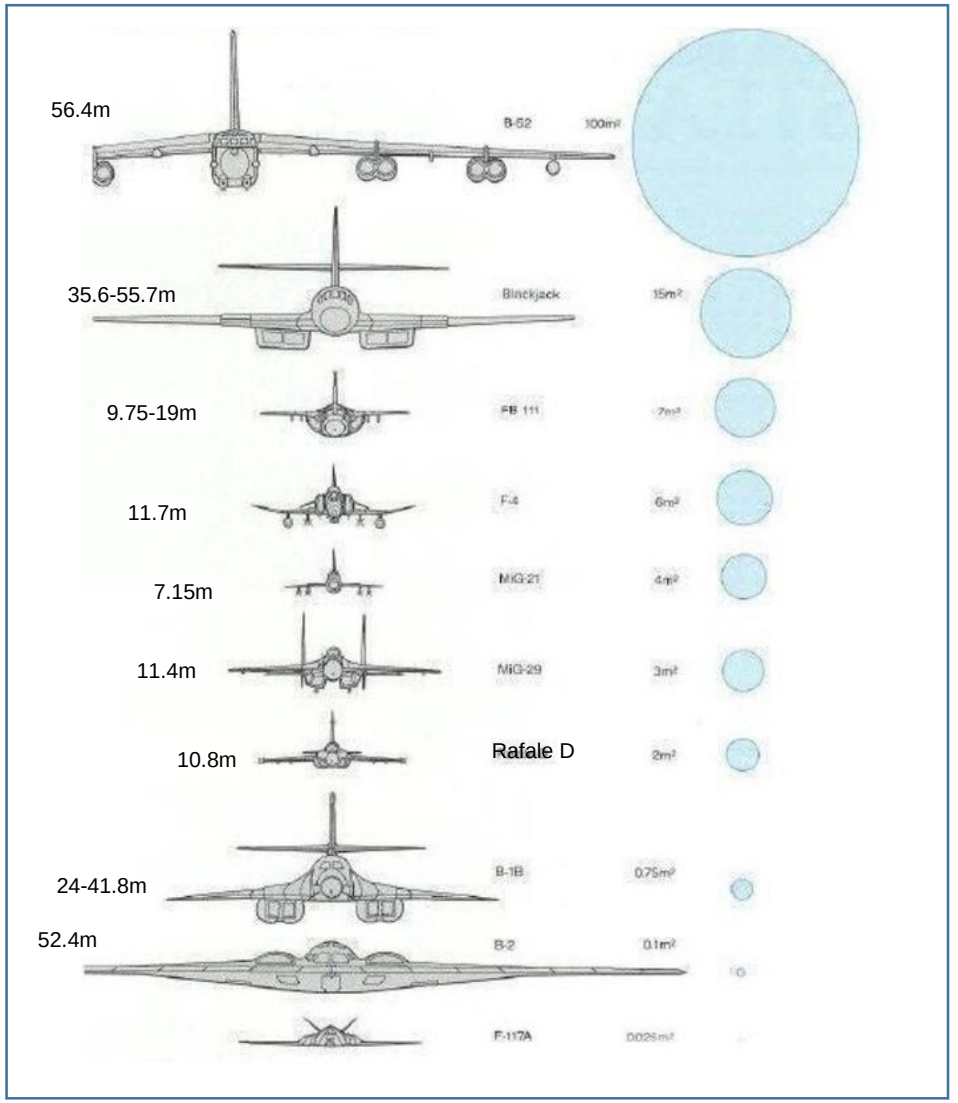


重量轻、可在相当宽微波波段(1 - 18 GHz)实现吸波。常用于微波暗室；采用角锥设计，以实现较高的反射损耗，通常应用于微波腔体、机壳、天线罩、网络机柜或天线等金属表面，是目前低成本吸波的主要解决方案。

隐形飞机用吸波涂层

- 隐形飞机





完美吸波的物理原理

- 电磁波入射到材料表面后，无反射与透射

$$R_{\mathrm{TE}} = |r_{\mathrm{TE}}|^2 = \left|\frac{\cos\theta - \mu_r^{-1}\sqrt{n^2 - (\sin\theta)^2}}{\cos\theta + \mu_r^{-1}\sqrt{n^2 - (\sin\theta)^2}}\right|$$

$$R_{\mathrm{TM}} = |r_{\mathrm{TM}}|^2 = \left|\frac{\varepsilon_r\cos\theta - \sqrt{n^2 - (\sin\theta)^2}}{\varepsilon_r\cos\theta + \sqrt{n^2 - (\sin\theta)^2}}\right|$$

$$A = 1 - R - T$$

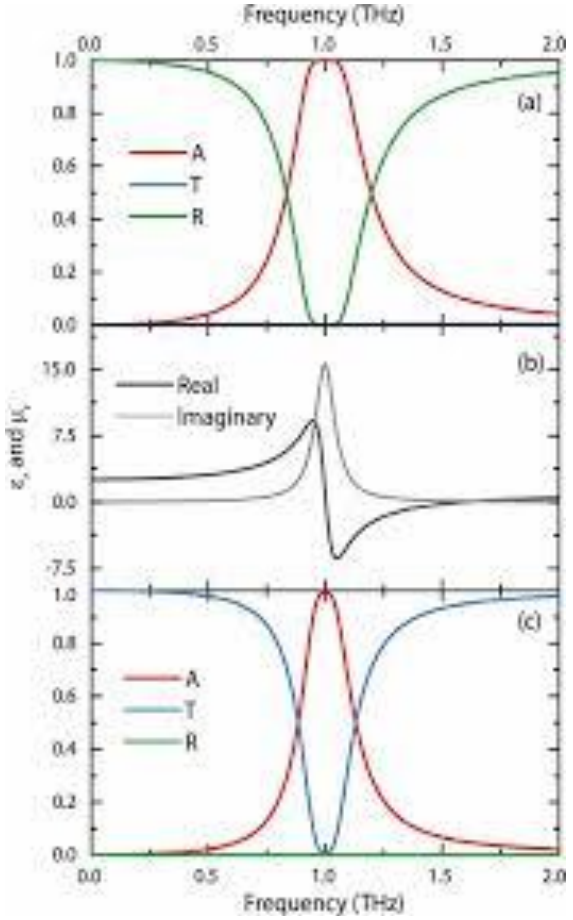
$$A = 1 - \left|\frac{Z - Z_0}{Z + Z_0}\right|^2 - T$$

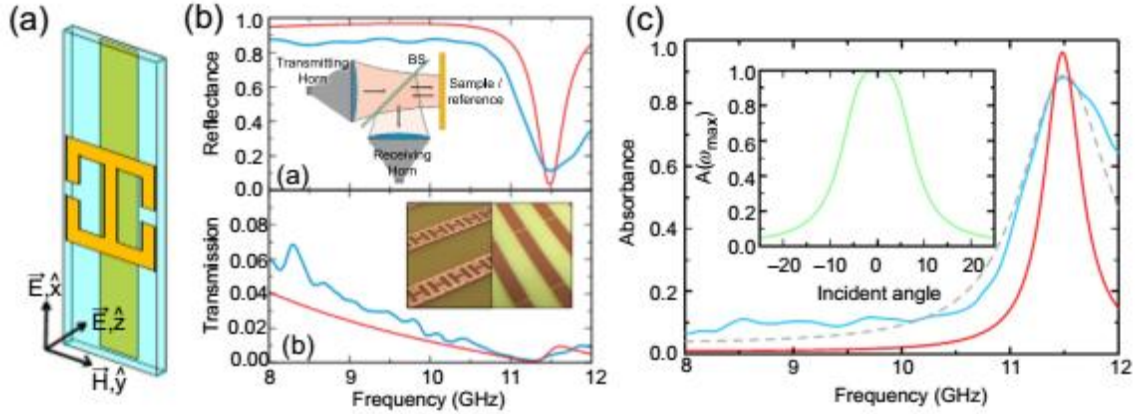
$$R_{\mathrm{TE}} = R_{\mathrm{TM}} = \left|\frac{\mu_r - n}{\mu_r + n}\right|^2 = \left|\frac{Z - Z_0}{Z + Z_0}\right|^2, \text{ where } Z = \sqrt{\frac{\mu}{\varepsilon}} \text{ and } Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

- 如何实现阻抗匹配？超材料

完美吸波超材料

- 通过超材料的结构分别实现特定的介电常数与磁导率，并使其相等

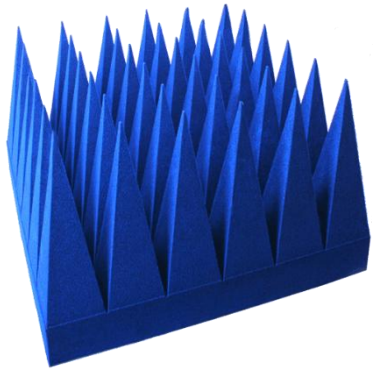




$\varepsilon=\mu$

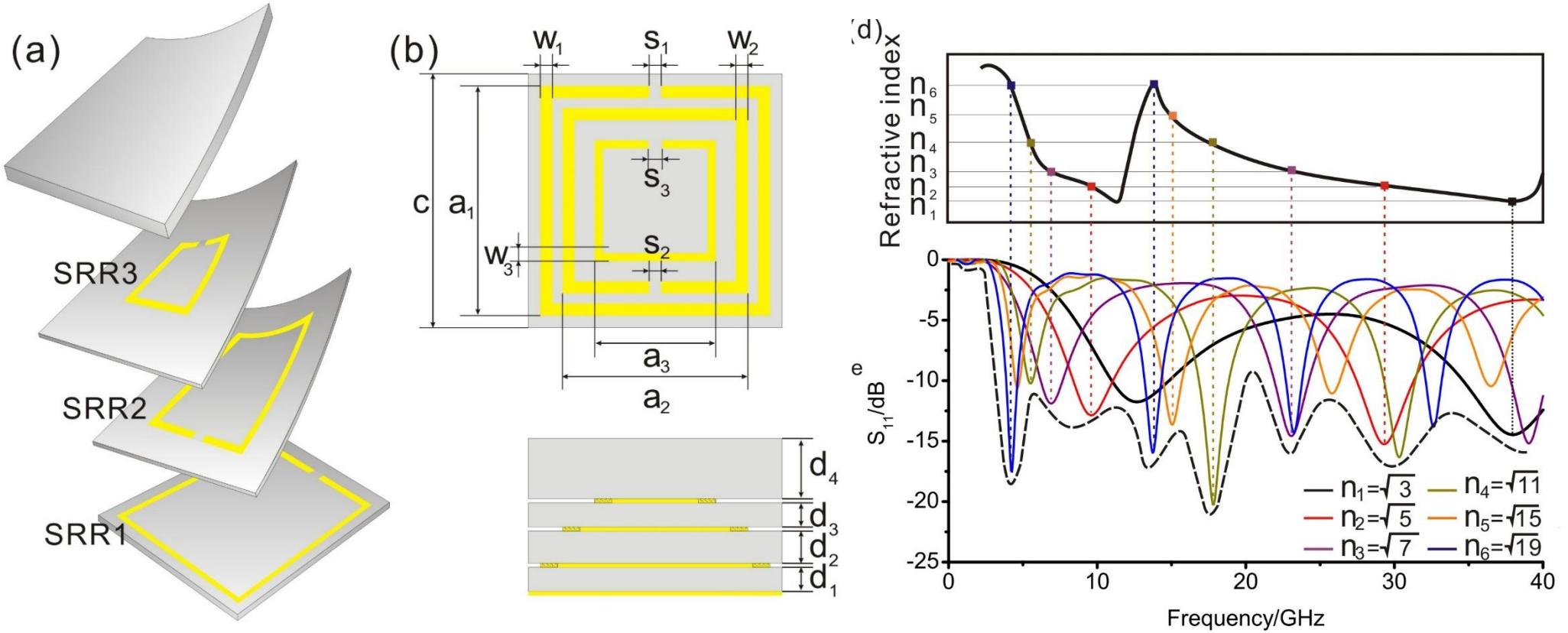
$Z=Z_0$

$$R_{\mathrm{TE}} = R_{\mathrm{TM}} = \left|\frac{\mu_r - n}{\mu_r + n}\right|^2 = \left|\frac{Z - Z_0}{Z + Z_0}\right|^2,$$



超宽带超材料

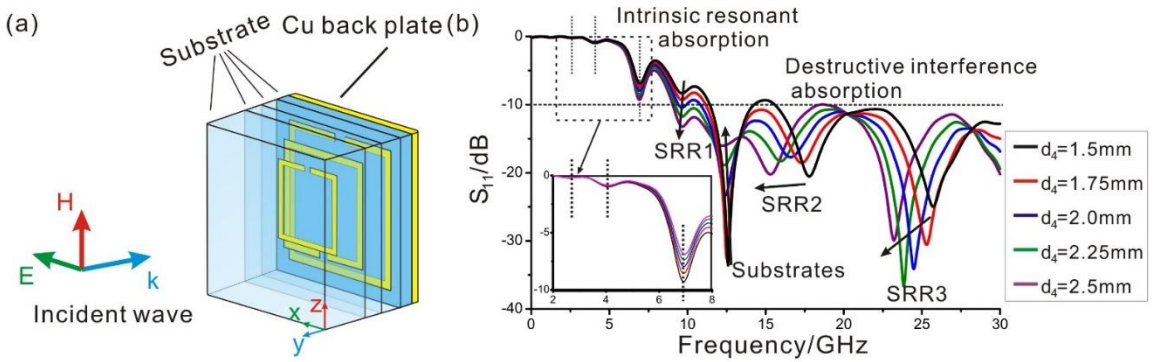
- 基于消反模实现宽带吸波

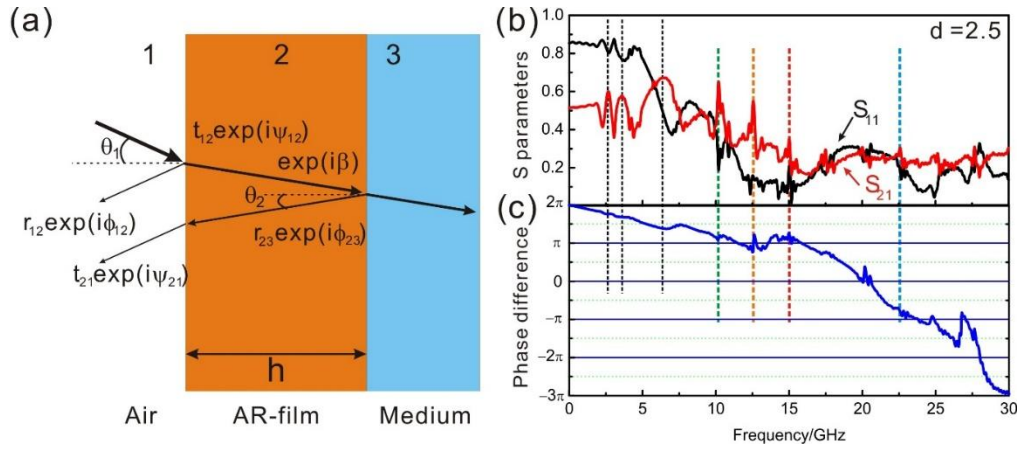


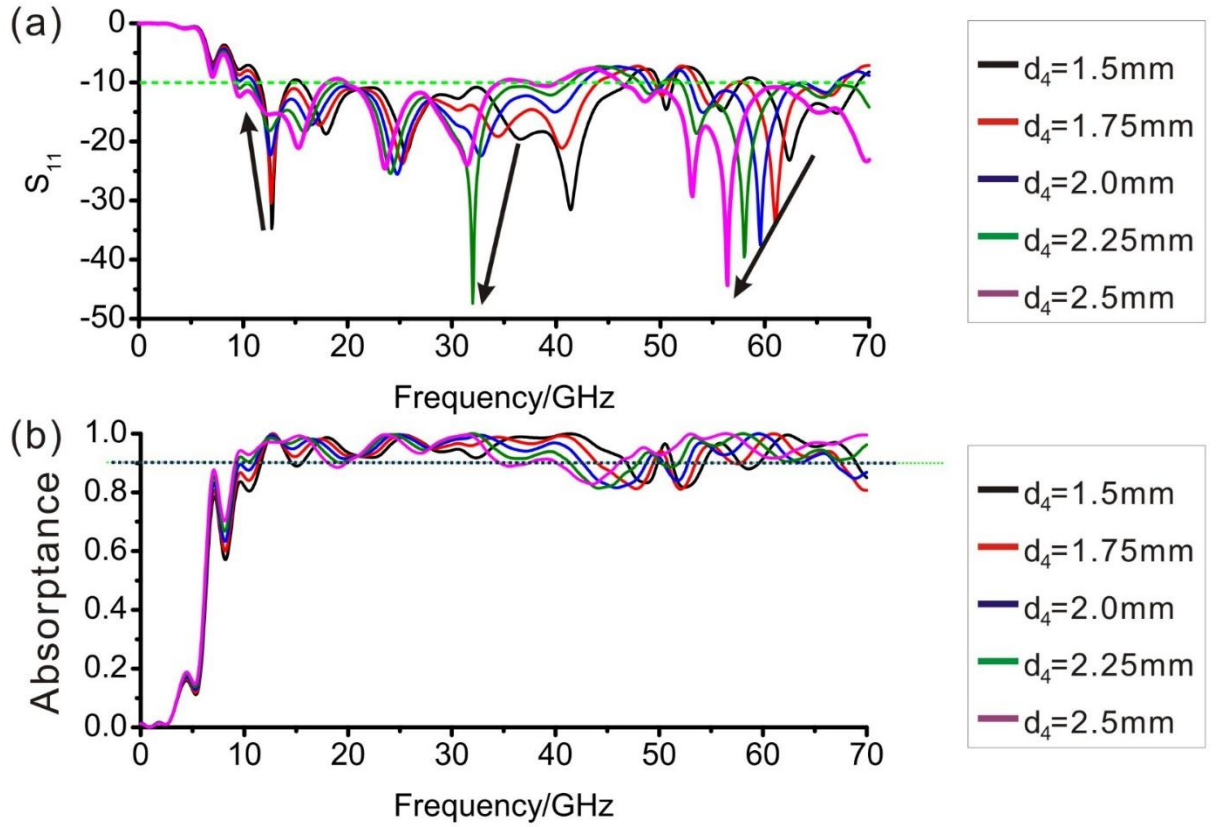
J Sun, et al. Optics Express 19 (22), 21155-21162, 2011

超宽带超材料

- 设计具有介电色散的超材料，在相当宽的频段内满足消反条件



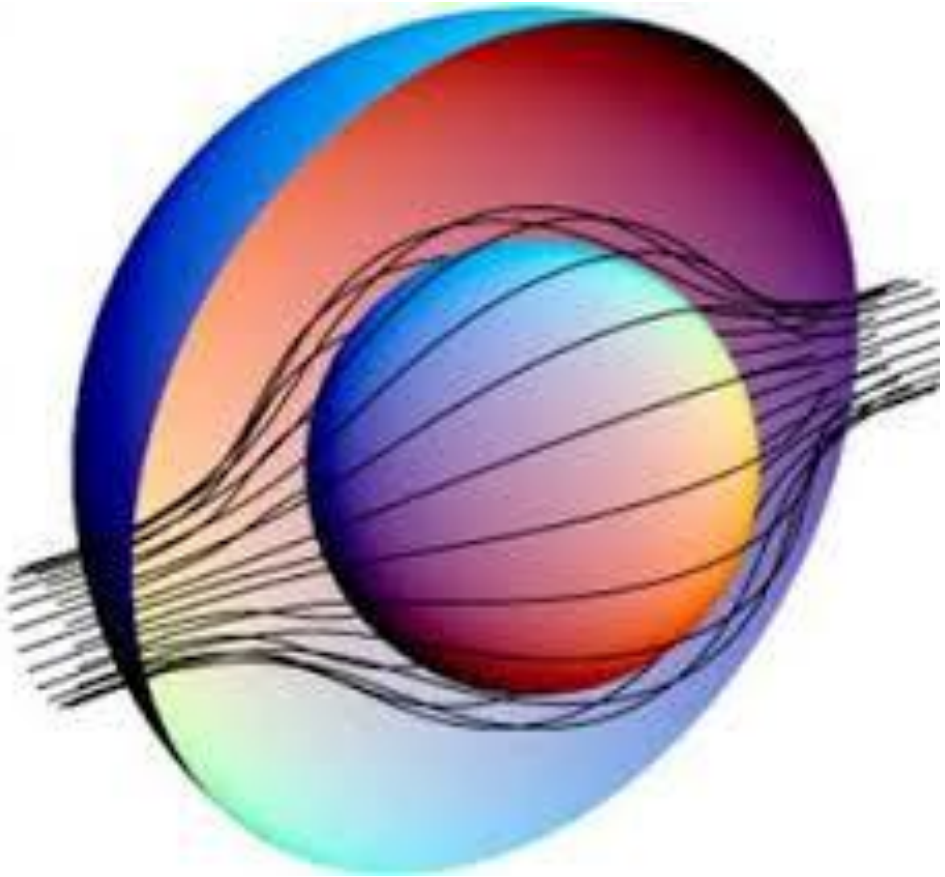




隐身衣

- 什么是隐身?

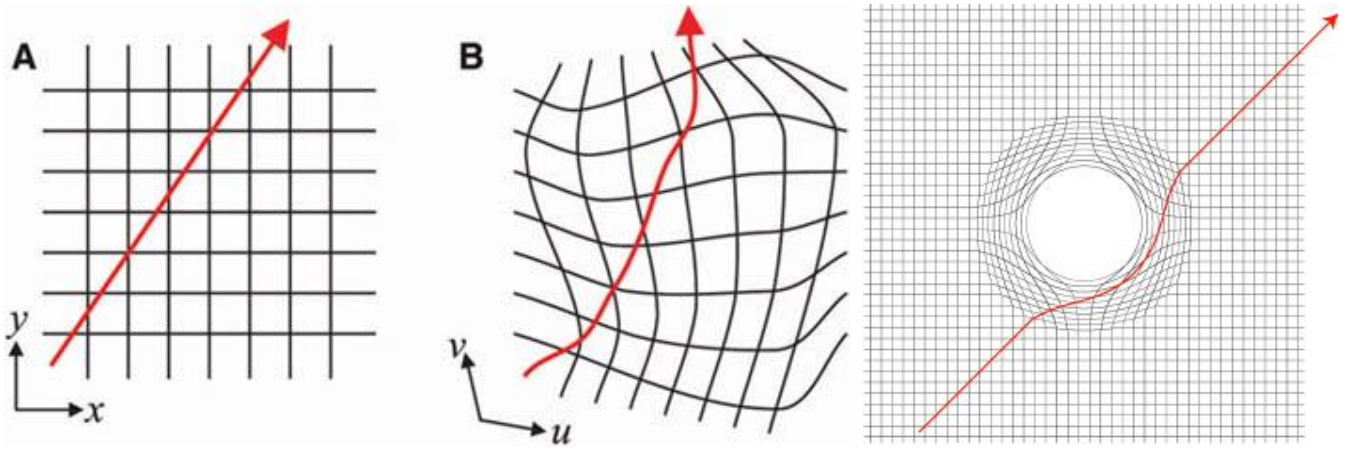




变换光学

- 通过材料实现数学上的空间变换，扭曲光线

  - Assume grid lines on a rubber sheet.
  - Grid lines can be distorted by stretching and squishing.
  - Light that followed a straight path in A, would follow curved path as in B.
  - Warping space to control the flow of light.

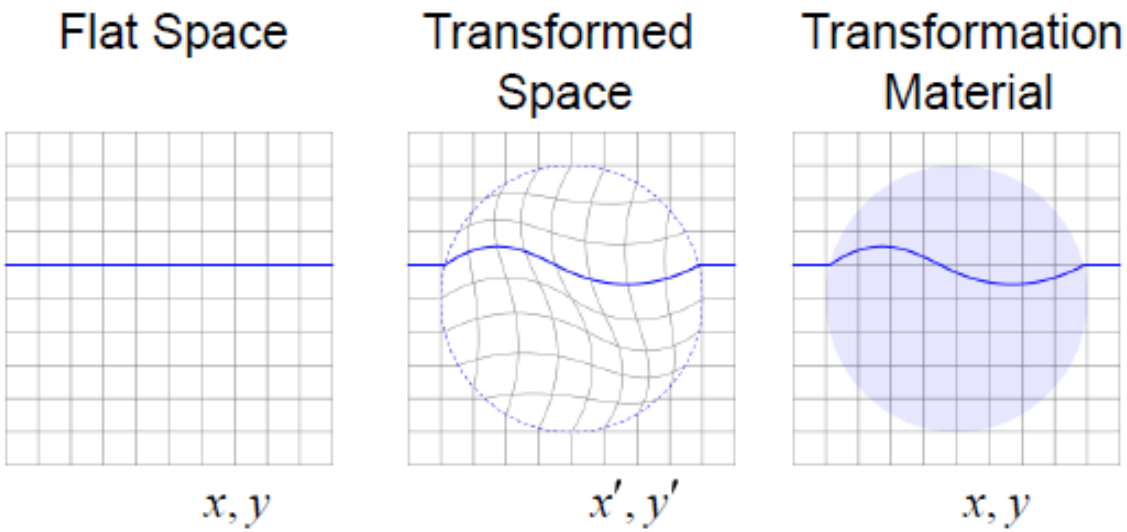


Left: A field line in free space with the background Cartesian coordinate grid shown. Right: the distorted field line with the background coordinates distorted in the same fashion.

变换光学

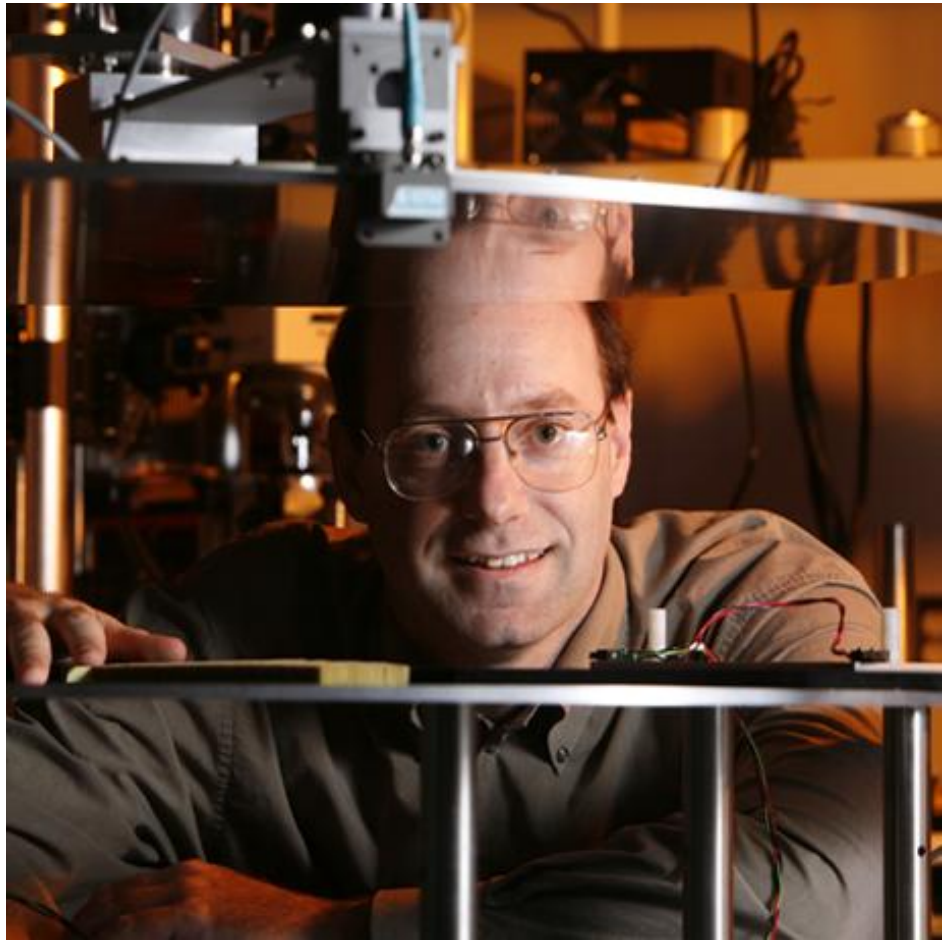
- 特定的折射率分布控制光的传输

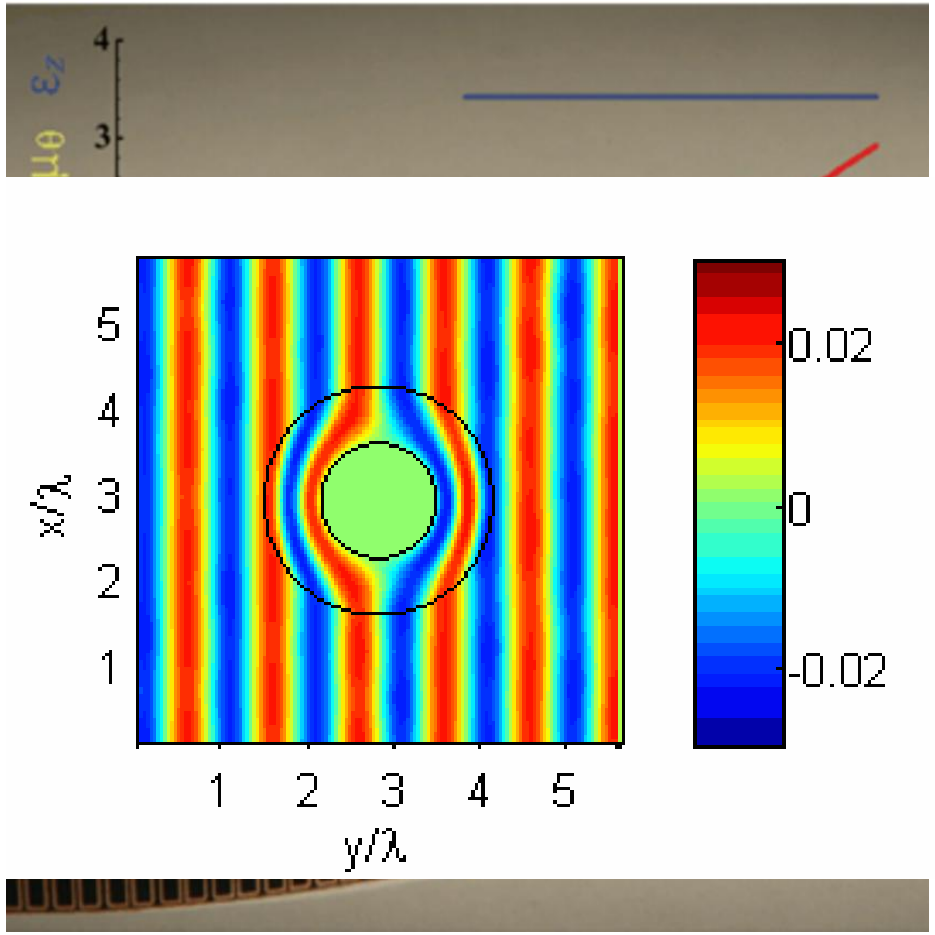
  - **Topological implementation**-the space should be deformed, assuming that material properties remain the same
  - **Material implementation**-the material properties are modified.
    - Material properties are generally varied by using anisotropic materials to guide the waves along a certain trajectory.
  - Thus, TO gives us an idea of what parameters to choose for the metamaterial device.



超材料隐身衣

- 利用开口环的几何尺寸实现任意磁导率，进而所需折射率

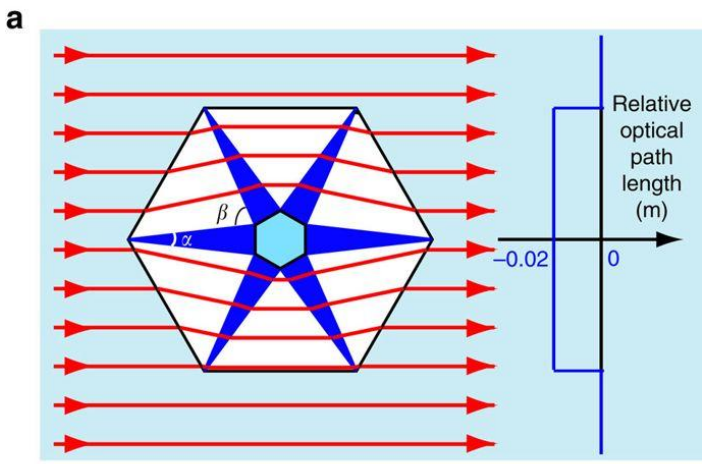


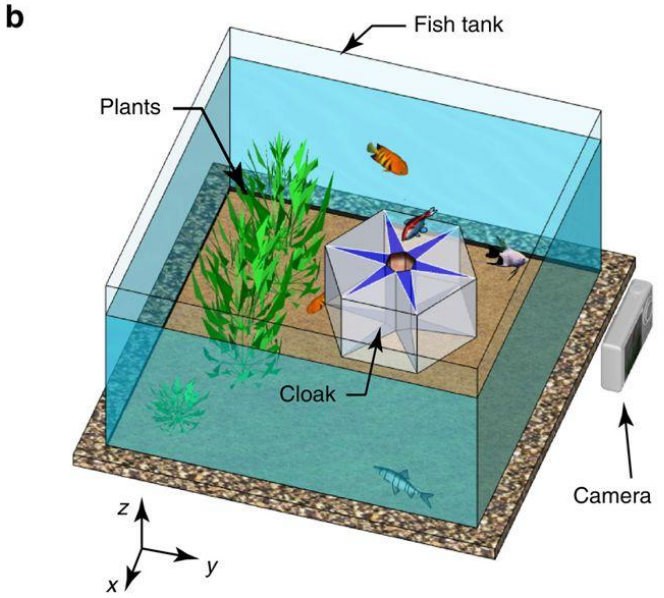


光学隐身






a
β
α
Relative optical path length (m)
–0.02
0

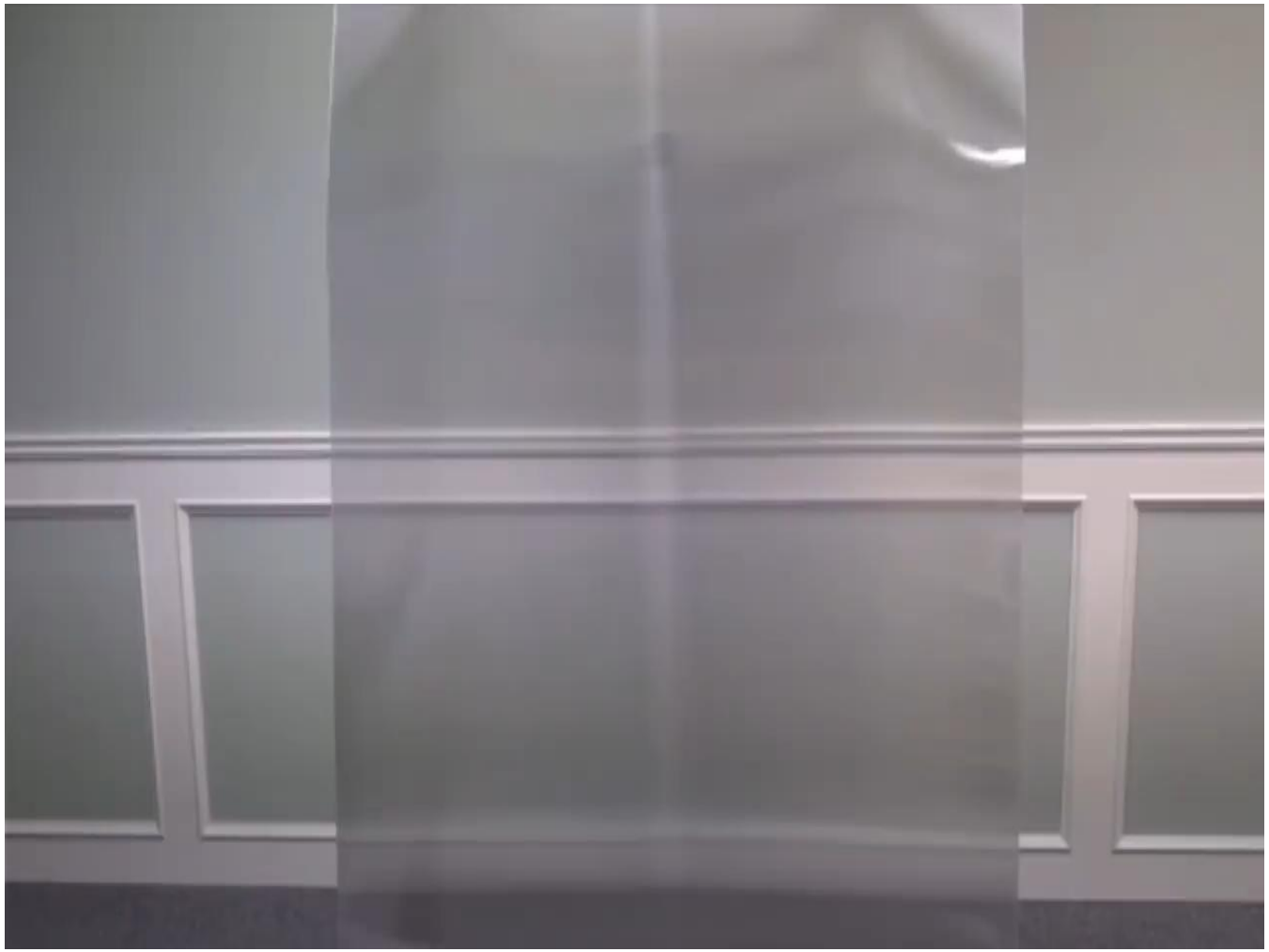

b
Fish tank
Plants
Cloak
Camera
z
y
x

源于生活



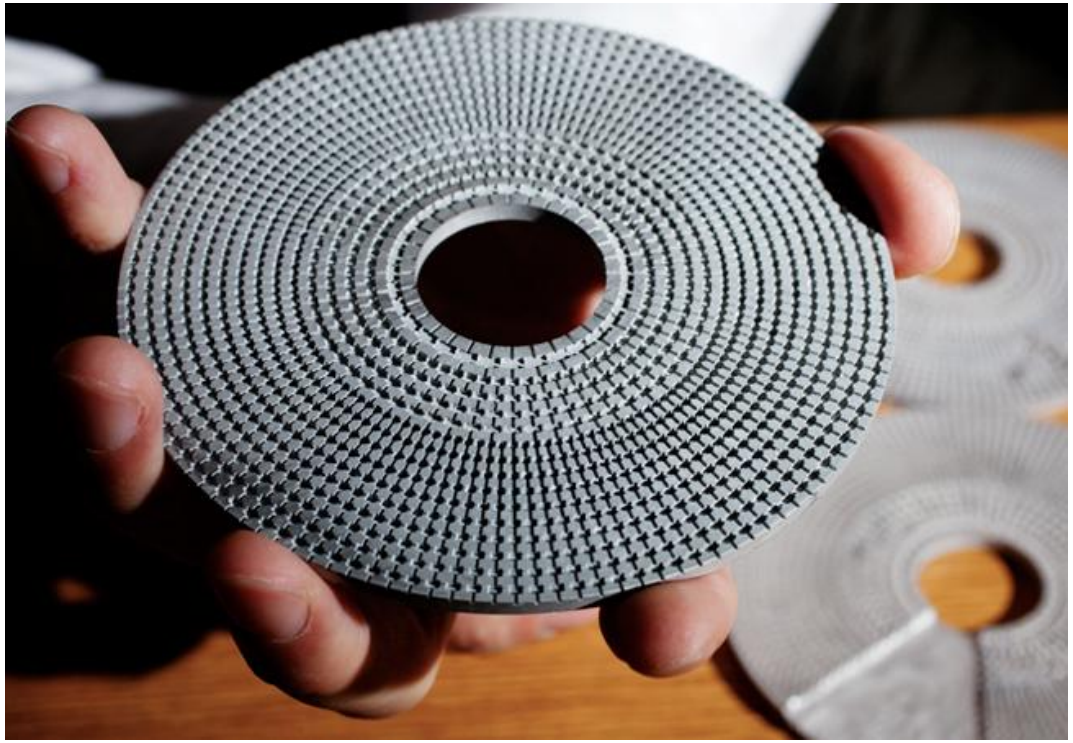


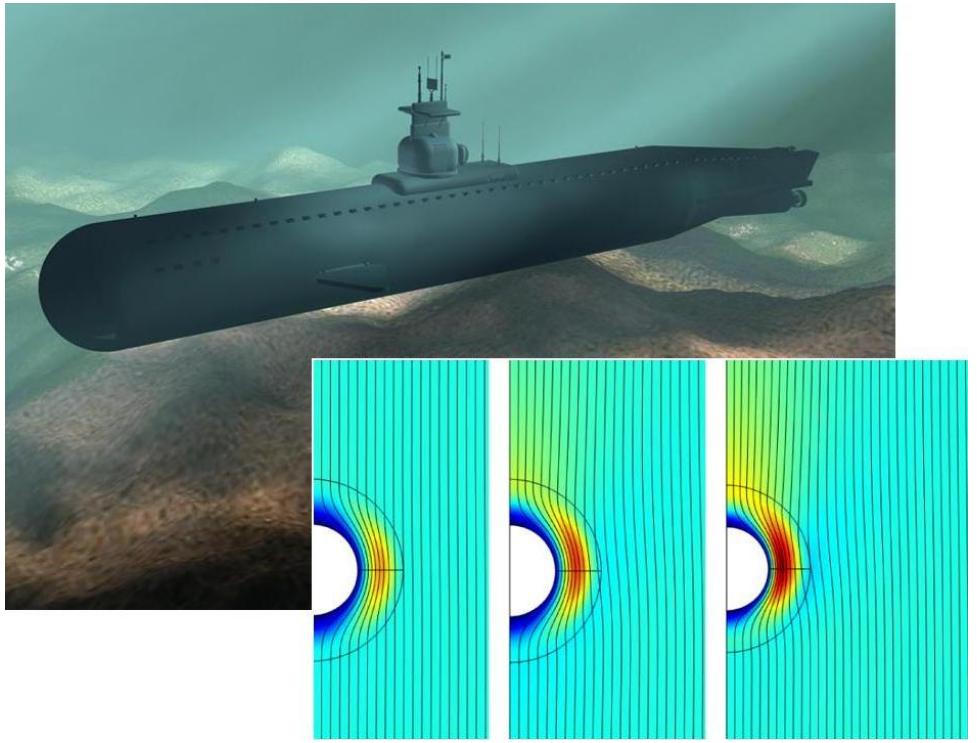
高于生活



其他超材料隐身衣

- 声学隐身

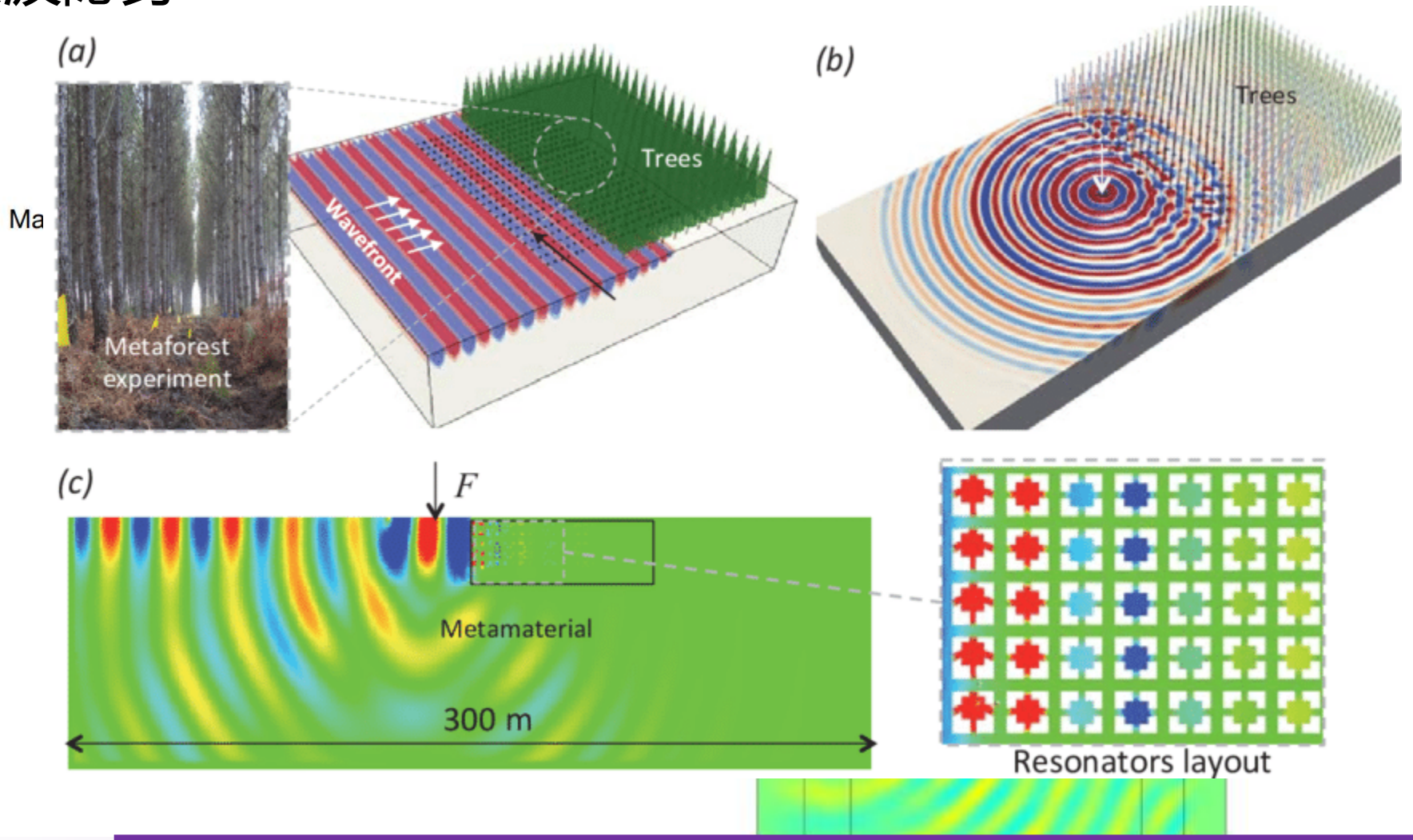




其他超材料隐身衣

- 地震波隐身

"Experiments on Seismic Metamaterials: Molding Surface Waves" 2014
Brule, et al, PRL 2014



通过超材料获得各向异性，突破衍射极限

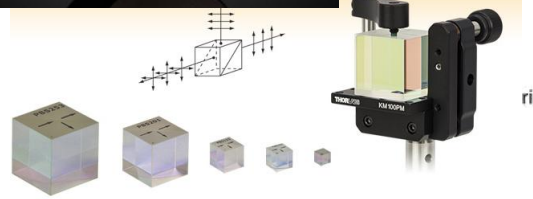
- 自然材料的各向异性

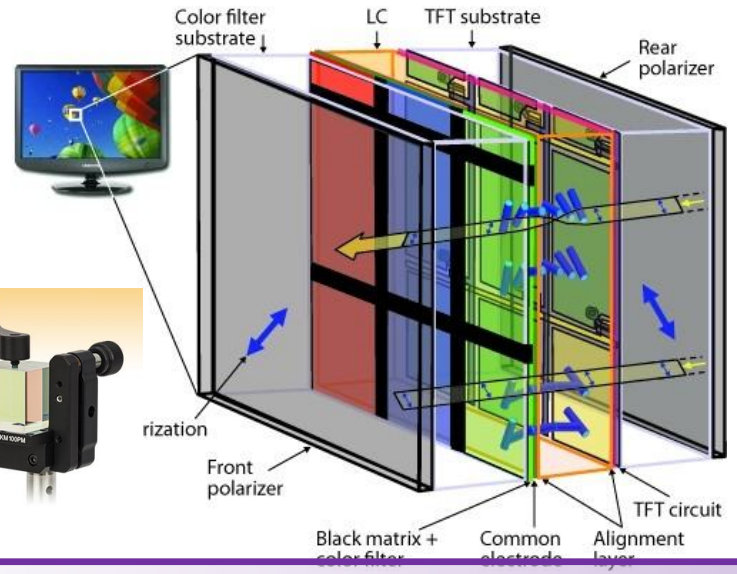


自然材料的各向异性：

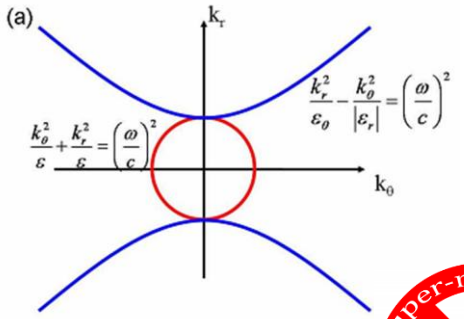
- 各向异性微弱
- 处于直角正交体系

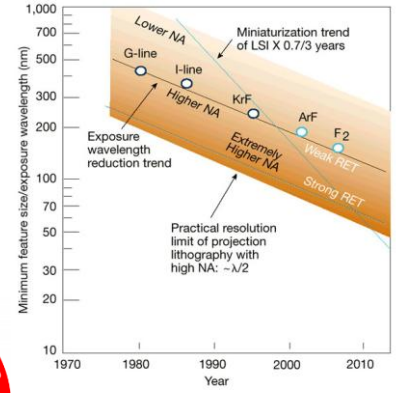






超材料的各向异性：双曲各向异性







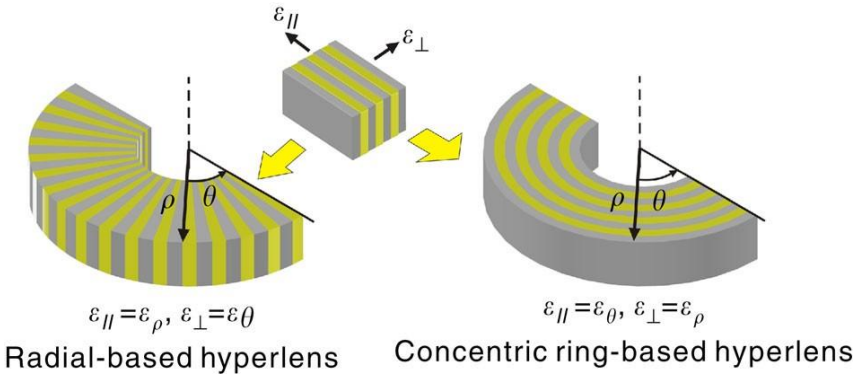


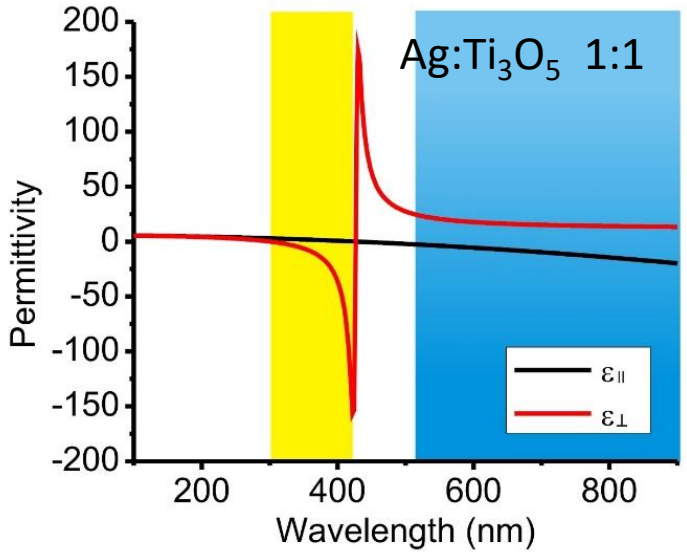
VS

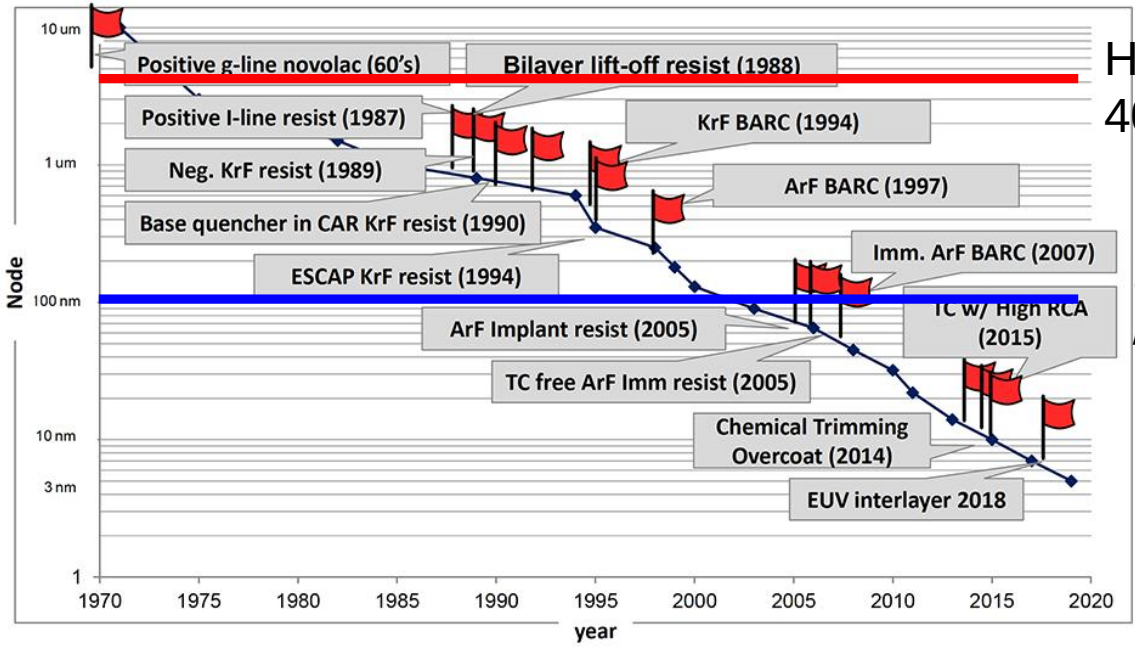


双曲超材料及其应用

- 双曲超透镜



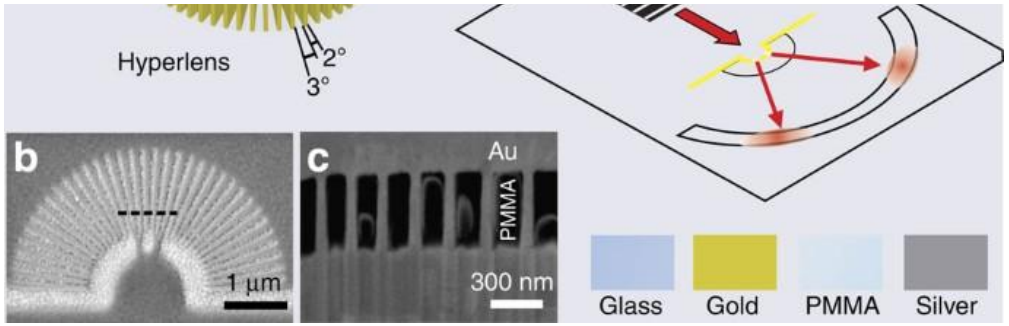




H line 405nm

248 /193nm

BARC = Bottom Anti-Reflective Coating
TC = Top Coat
RCA =Receding Contact Angle
Imm. = immersion



超材料的未来发展方向

- 超材料的发展方向

工程化应用


Engineering 17 (2022) 1–2

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Engineering

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Editorial

Metamaterials: From Engineered Materials to Engineering Materials

John Pendry [a], Ji Zhou [b], Jingbo Sun [b]

[a] Department of Physics, Imperial College London, London SW7 2AZ, UK
[b] School of Materials Science and Engineering, Tsinghua University, Beijing 100084, China

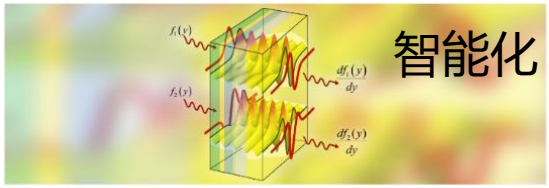
John Pendry Ji Zhou Jingbo Sun

As a new frontier in materials science, metamaterials have experienced a boom in the past two decades. This research field began with artificially structured materials exhibiting extraordinary electromagnetic properties unavailable in nature; now, it extends to a large family of artificial-material systems with wide-ranging, extraordinary, and outstanding properties for applications in fields such as electromagnetics, optics, mechanics, thermodynamics, and acoustics, based on engineered functional units instead of inherent characteristics. The focus of metamaterial research has eventually shifted from new principles, new phenomena, and new structures to new properties and new functions for technological applications. Metamaterials are increasingly anticipated to provide technological solutions to many challenges in engineering. The nature of metamaterials is transforming through an evolutionary process from engineered materials to engineering materials. This special issue reflects on some clues to this historical process.

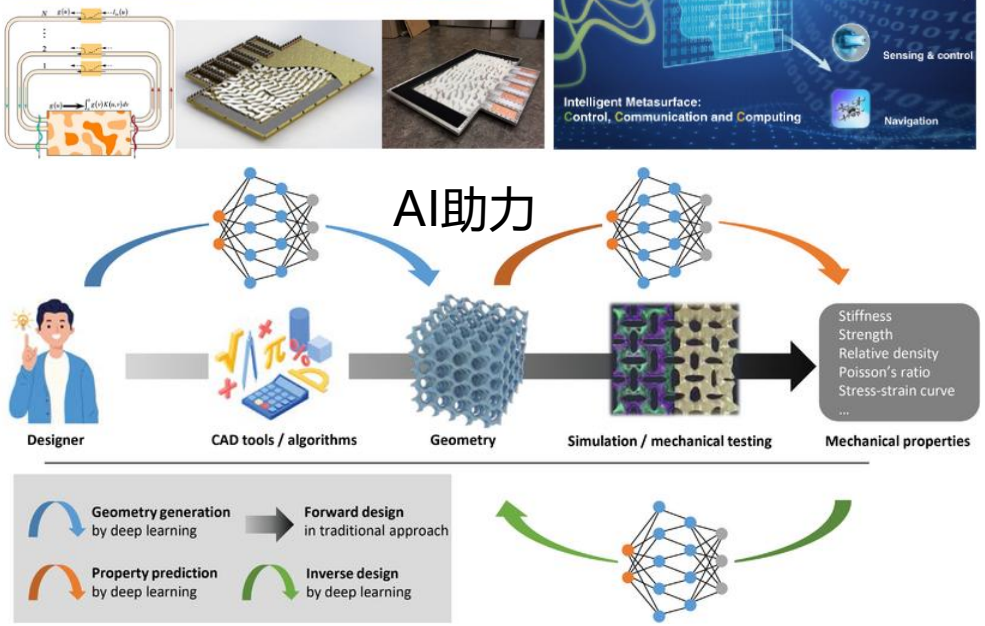
Inspired by the realization of negative-index materials, the first metamaterial was born at the end of the 20th century. Two significant inventions by Sir John Pendry took the lead in achieving supernatural properties artificially. In the first of these inventions, metal wires exhibited an effective negative permittivity at a low plasma frequency; in the second, a split ring resonator mimicked the circulating current in a molecule, enabling the generation of artificial magnetism. These inventions directly paved the way for the achievement of a negative index with negative permittivity and permeability; they also helped in developing the idea of achieving physical properties beyond those present in nature through artificial structures with engineered units. Pendry's illustration of negative refraction, but also enlightened researchers on the fact that there was a method to artificially manipulate permeability. In the same way, Pendry's work on the refractive index brought in a series of novelties. The negative index itself can be used as a perfect lens that can break the diffraction limit in microscope imaging. In addition, a metamaterial with a desired refractive index distribution can guide light beams to pass around a central region that is thereby rendered invisible to the light, in what is called "cloaking." Other achievements, such as the metamaterial absorber, zero-index material, and electrical-induced transparency can also be attributed as the designable ε and μ. These works emerged during the ten-year period immediately after metamaterials were first proposed. During that decade, metamaterials flourished in electromagnetics and expanded to optics as well.

With the success of this concept, properties beyond those that appear in nature are not limited to electromagnetics or optics. Metamaterials are also boosting innovations in mechanics, acoustics, and thermodynamics, pushing new boundaries in the entire realm of physics. During the last 20 years of progress, the design ideology and fabrication techniques for metamaterials have been well developed. From a scientific perspective, metamaterials supply a very powerful platform to nourish new physics; from an engineering perspective, they are now a well-established tool, ready for real applications. Researchers are paying increasing attention to functionalities that can significantly improve current technologies; thus, these engineered materials are transitioning from scientific research to the field of engineering. In this special issue, we introduce recent achievements from several top research groups in the field of metamaterials to report on promising applications of metamaterials and metamaterial-based devices in engineering.

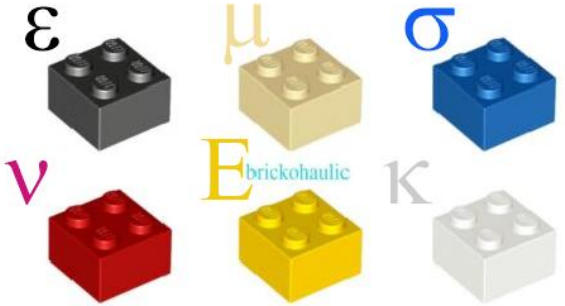
Microwave metamaterials have been the most extensively studied metamaterials, since they were the first realization of the concept of a metamaterial. Many new electromagnetic effects were also demonstrated for the first time in the microwave range, mainly to obtain material properties that do not show up at high frequencies or to avoid fabrication issues from nanofabrication. Nowadays, microwave metamaterials are also the closest in the metamaterials family to achieving real applications and being used industrially, especially in communication engineering. In this

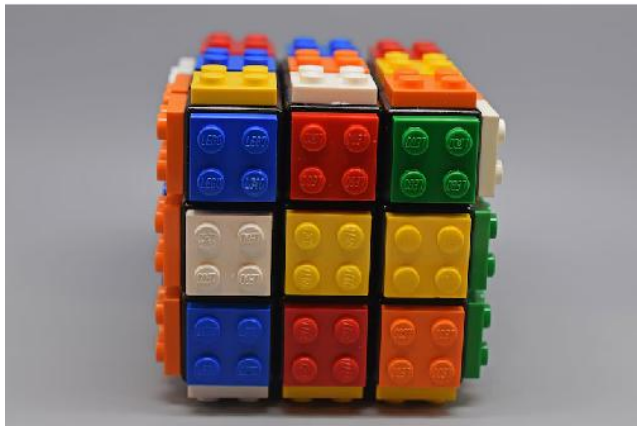






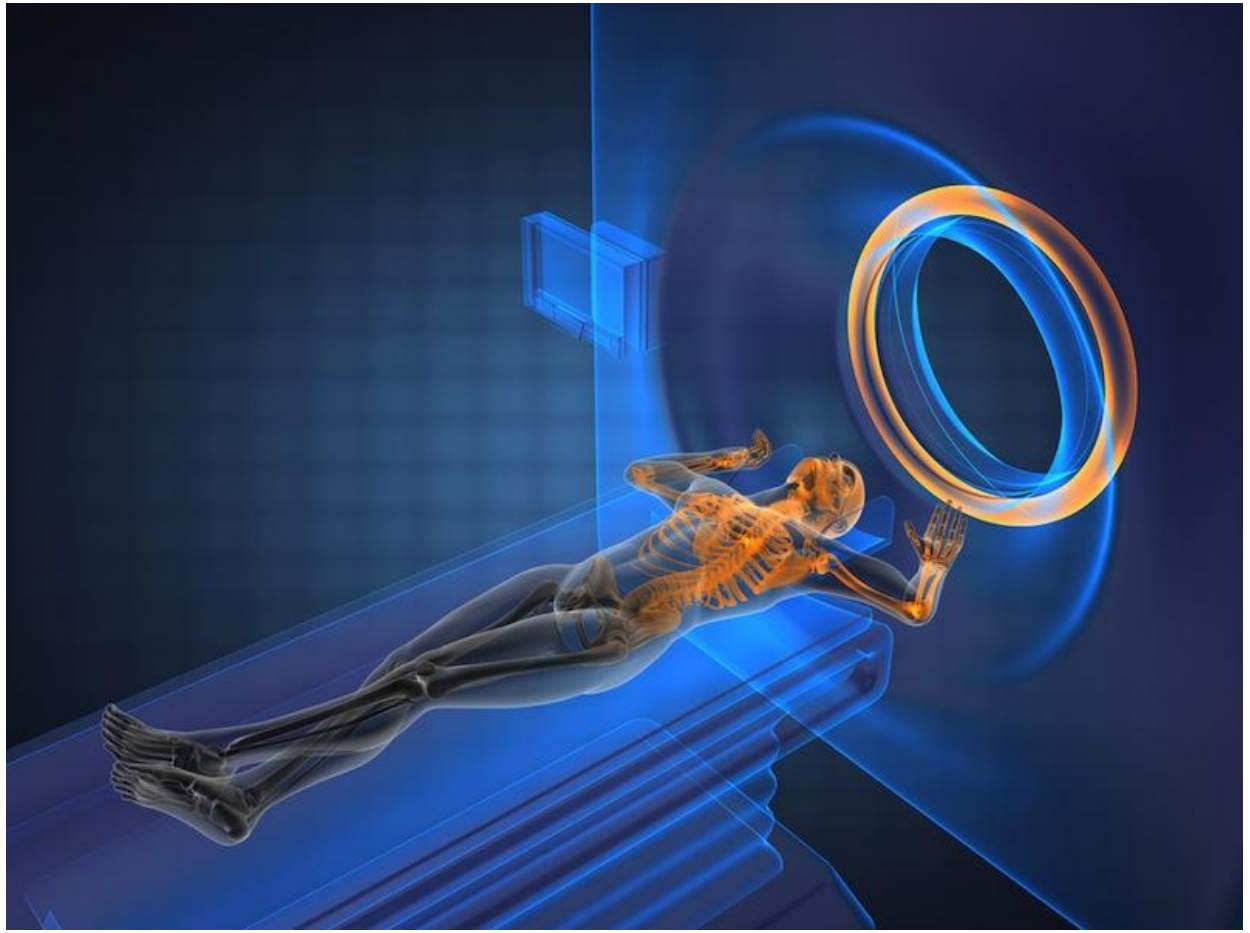
多物理场的解耦耦合





超材料的应用

- 超材料在医学影像领域的应用—核磁线圈







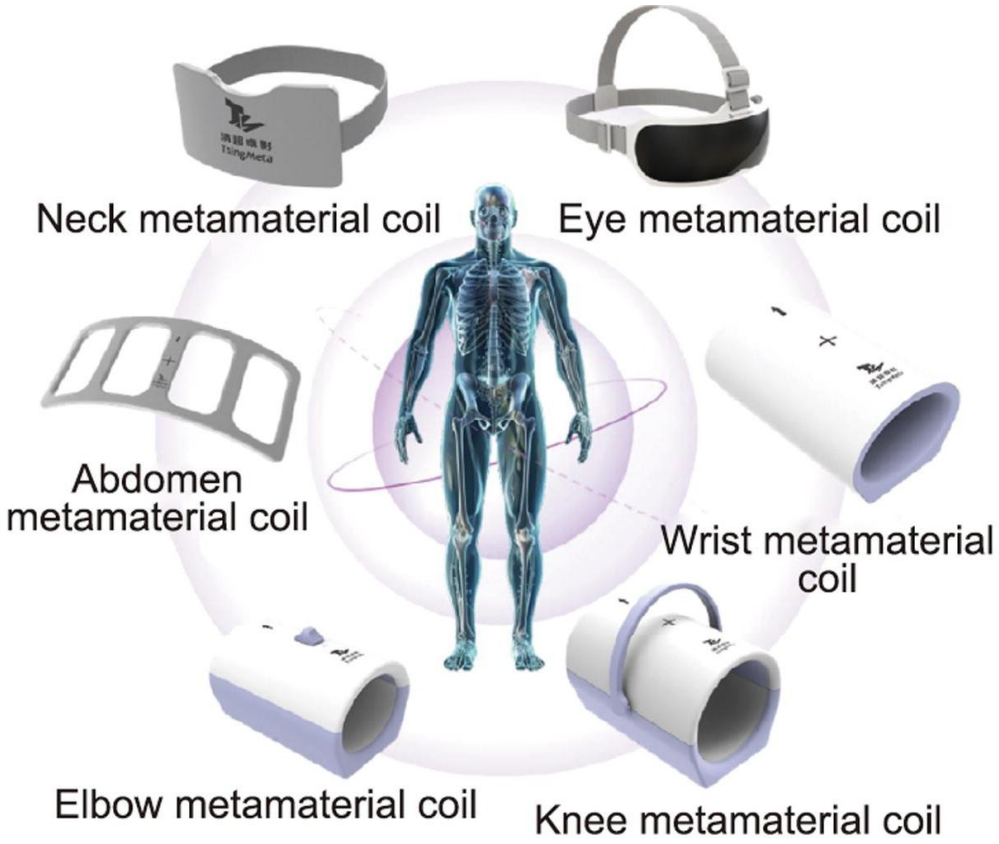
核磁成像是医学影像的重要手段，与x射线相比，损伤小、软组织分辨率高

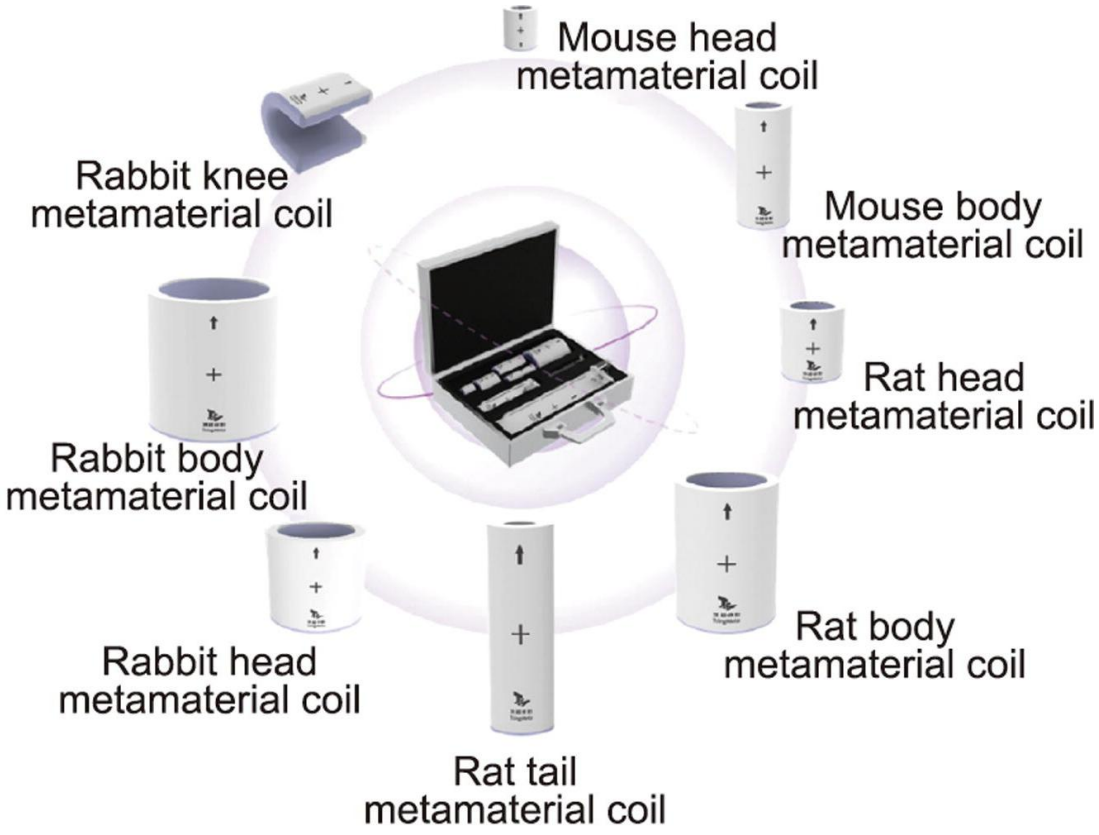
核磁成像的分辨率直接决定于施加磁场的场强与扫描时间。

核磁成像设备通常有1.5T、3T、5T、7T等类型。

用于核磁成像的超材料线圈

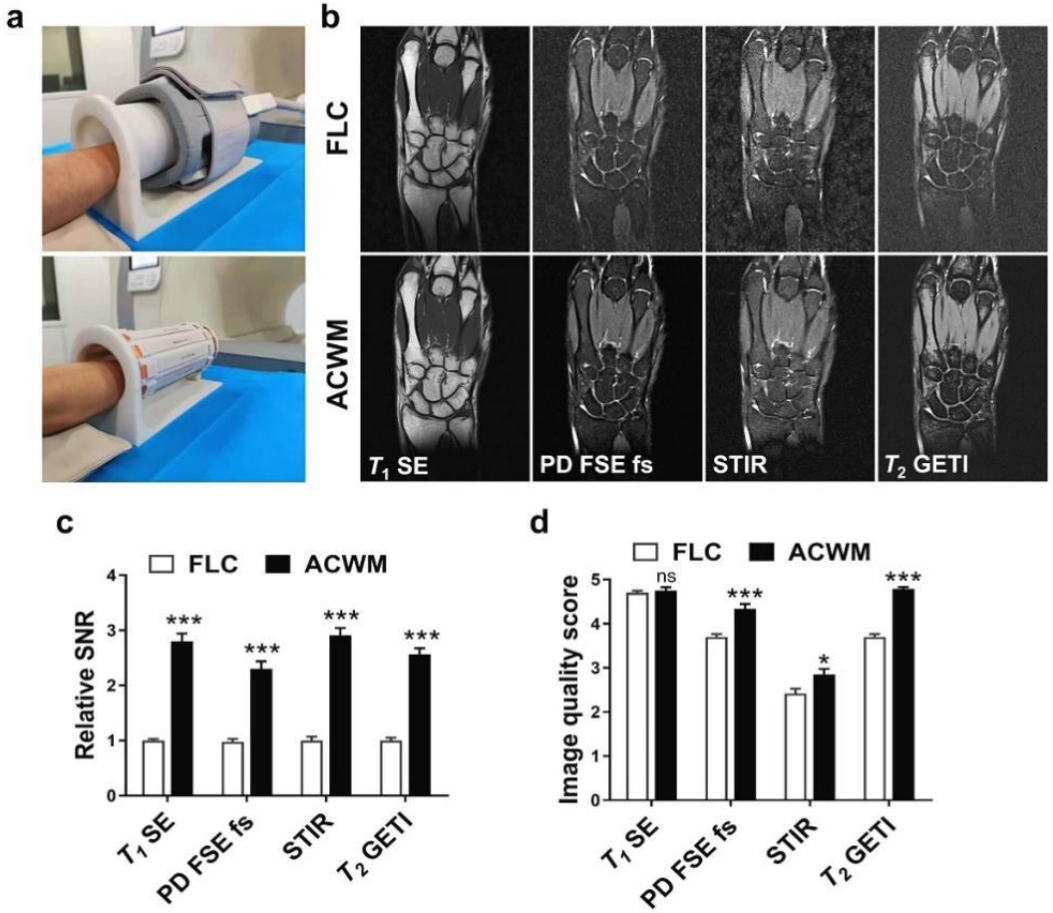
- 通过超材料设计实现高均一度磁场分布

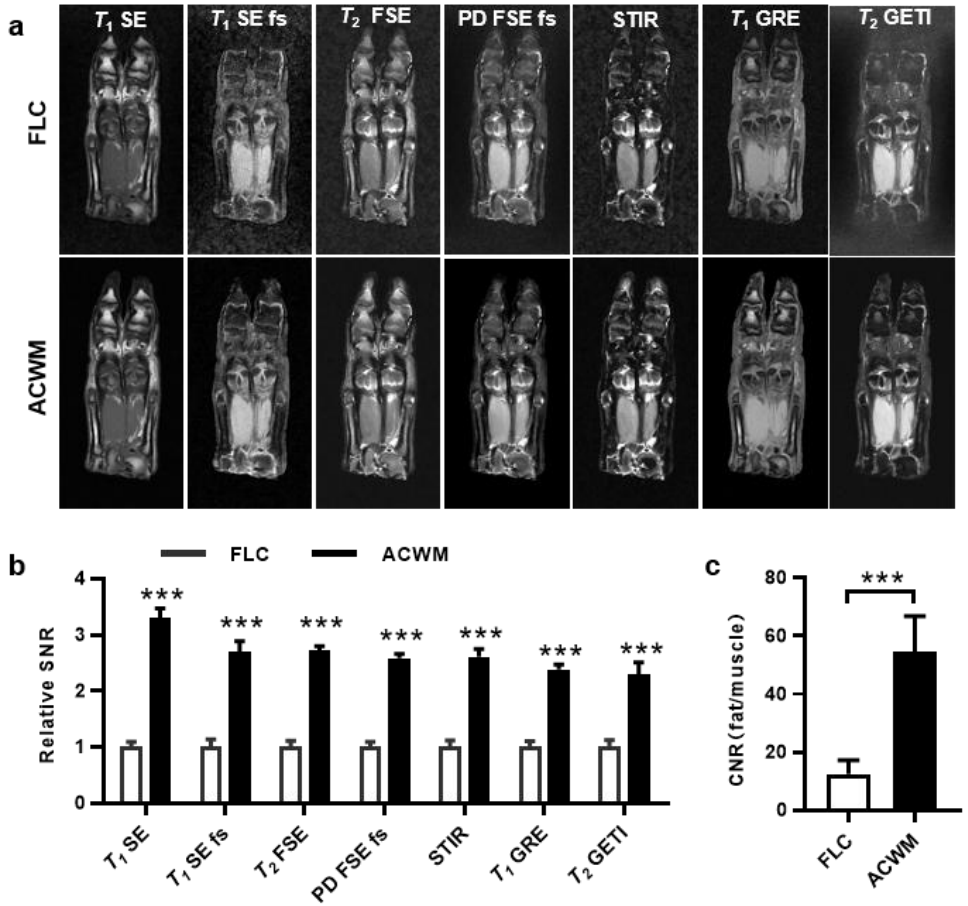




用于核磁成像的超材料线圈

- 分辨率提高2-4倍 **磁场： 1.5 T**

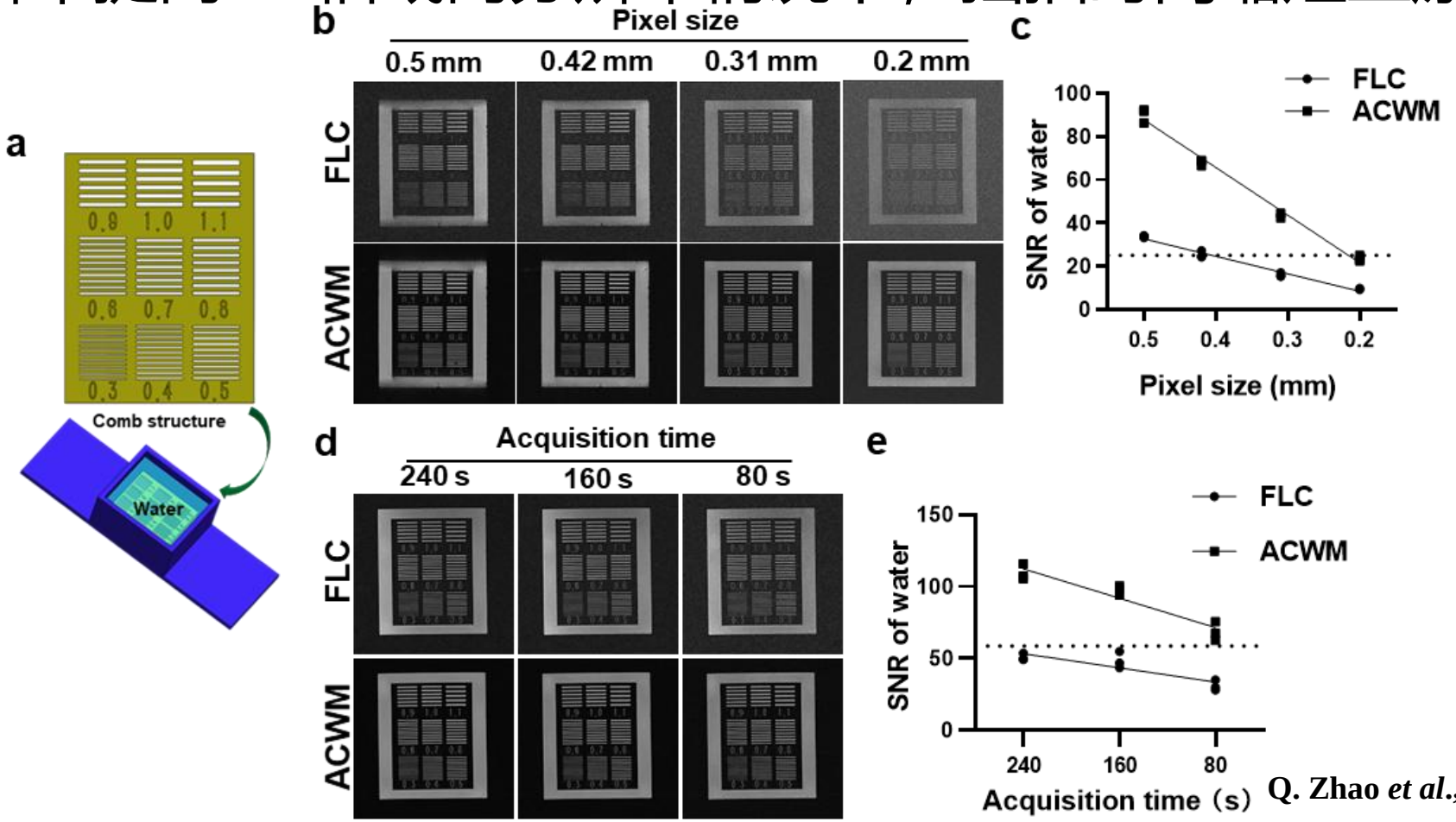




用于核磁成像的超材料线圈

- 分辨率提高2-4倍或同分辨率情况下，扫描时间缩短至原来的1/3

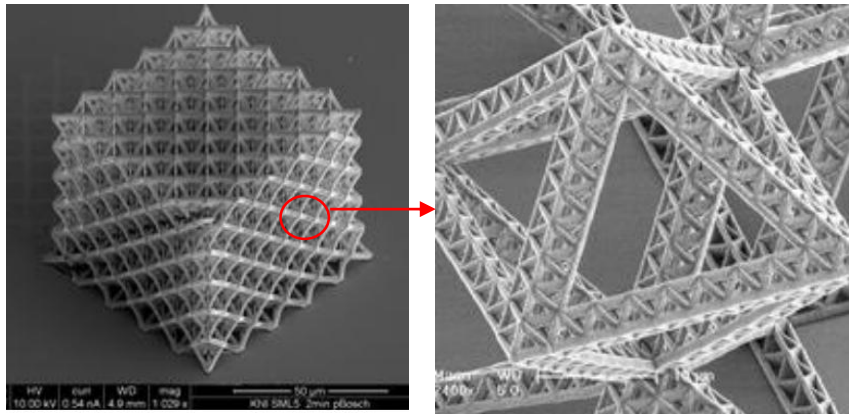
磁场： 1.5 T

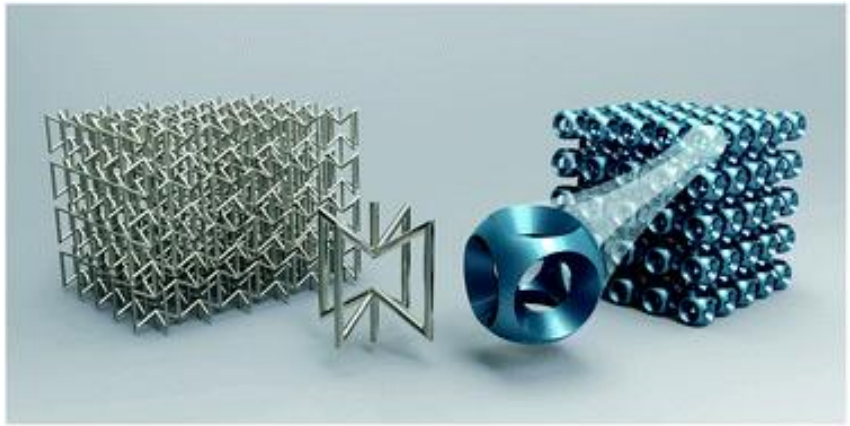


Q. Zhao *et al.*, Adv. Mater. 2021, 202102469

材料科学中的艺术—超材料

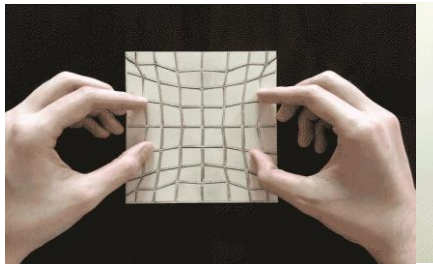
- 形

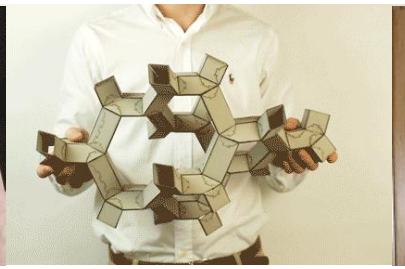




- 神
  - 富有想象力与设计性，由材料构成，超于常规材料。
  - 超材料的研究无拘无束、充分发挥人的创造性。

    无中生有、无与论比、无所不能
  - 超材料与艺术是相互融合的







  - 从艺术品到工艺品

期待交流，谢谢！

人才培养—本科生培养

◈积极融入本科生书院制培养

- 2020年 首批进入“**强基计划**”本科生培养（未央书院）
- 2022年 首批进入“**为先书院**”本科生培养（为先书院）
- 2025年 首批进入“**自强书院**”本科生培养（自强书院）

WEIYANG COLLEGE TSINGHUA UNIVERSITY
清华大学未央书院

为先书院

强学生之基、强学科/学校之基、强国家之基

成人成才、通专融合、本博贯通

激发学术志趣、投身国家关键领域

以培养工科国际顶尖学术、产业、管理人才为先

以发展中国工业核心关键技术为先

以建设清华工科创新人才本科培养示范为先

研究生招生

学术博士（推免，50名左右）：

- **材料科学与工程　一级学科　（工学）**

  二级学科：材料学、材料物理与化学 、材料加工工程

  学科方向：金属材料、材料科学、陶瓷材料、材料加工

工程博士/硕士（全日制+非全日制）：

- **工程硕博士专项（材料与化工，30名左右）** 每年9月左右，由学院和联合培养企业共同组织综合考核（博士、硕士）
- **国家重点实验室工程博士　（10名左右）**
- **工程硕士 (推免/统招，20名)**
- **创新领军工程博士　（10名左右）**

师资队伍

教职工	98
教师	80
教授/研究员	41
副教授/副研究员	35
助理教授/助理研究员	4
高级工程师	12
工程师	6
博士后	100+
合同制人员	90+

两院院士	7
“长江学者奖励计划”特聘教授	10
国家杰出青年基金获得者	14
973/国家重点研发首席科学家	10
中组部千人计划教授	5
“长江学者奖励计划”青年学者	5
优秀青年基金获得者	12
中组部“青年千人”	10

基金委基础科学中心：1　　基金委创新群体：1

科技部创新团队：2　　教育部创新团队：2


朱静


李龙土


柳百成


柳百新


翁宇庆


南策文


周济

迈入顶尖前列材料学科思考

- 一流的师资队伍，国际学术影响力强！ “师资队伍国际化”
- 世界前沿原创性成果！ “引领者”
- 对国家重大工程、国民经济主战场支撑！ “国家需求牵引-清华标签”
- 多学科之间真正交叉融合！ “评价机制、体制”

家国情怀、目标牵引、勇于担当、乐于创新！



群聊: 2024级【未央-材料】
兴趣小组

该二维码7天内(11月17日前)有效，重新进入将更新

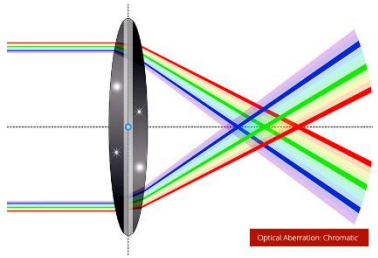


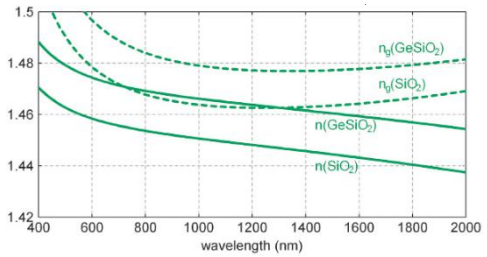
群聊: 2025级【未央-材料】
兴趣小组

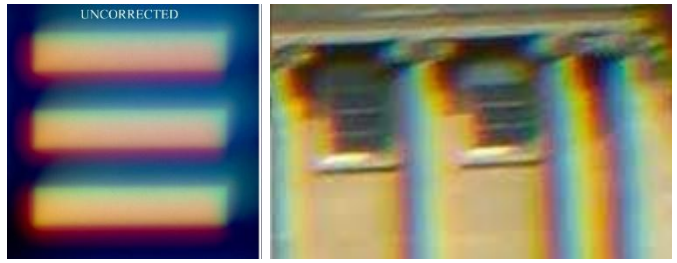
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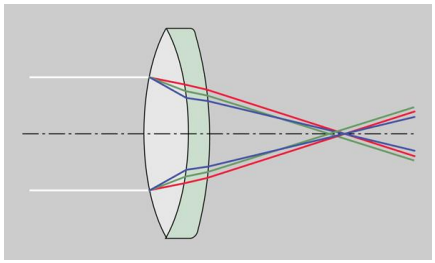
Metasurface application I: Metalens

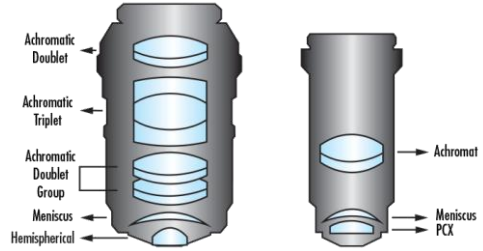
- Optical aberration of optical lens



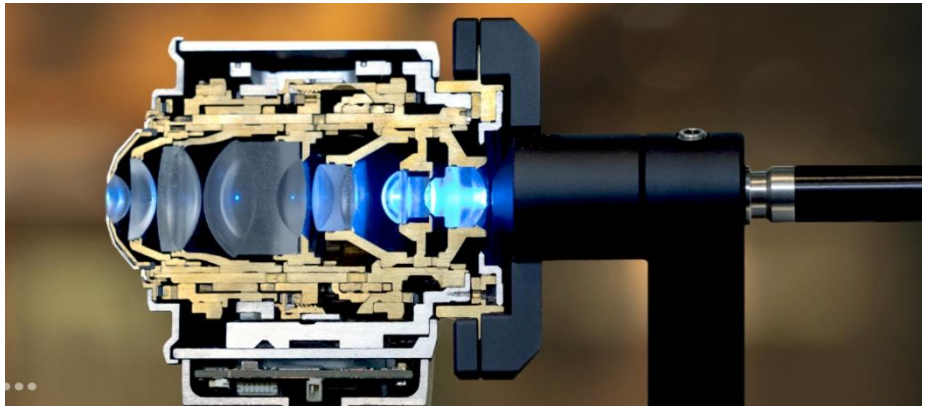






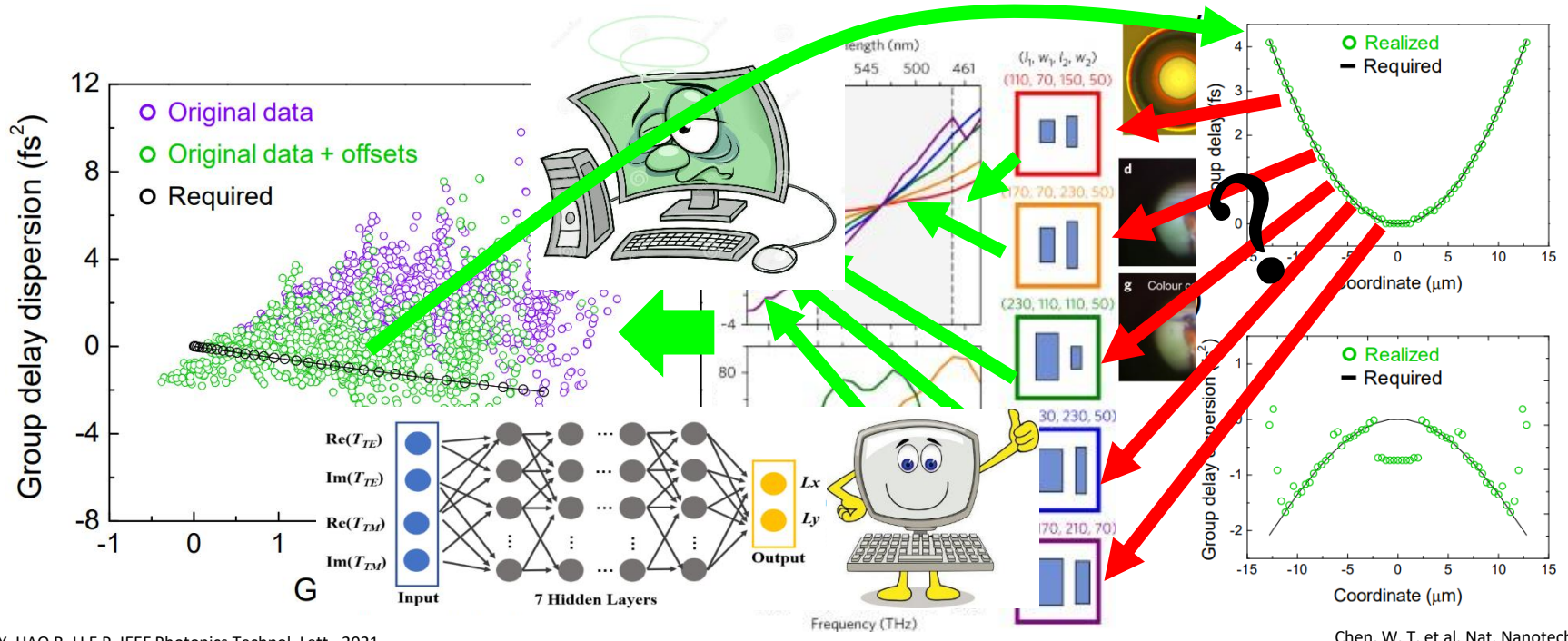






Challenge

- **Everything is clear. The performance of the lenses is excellent!**
- **But tremendous work in the design.**



GU Y, HAO R, LI E P. IEEE Photonics Technol. Lett., 2021.
LIN K-F, HSIEH C-C, HSIN S-C, et al. Appl. Optics, 2019.
AN S, ZHENG B, TANG H, et al. Adv. Opt. Mater., 2021.

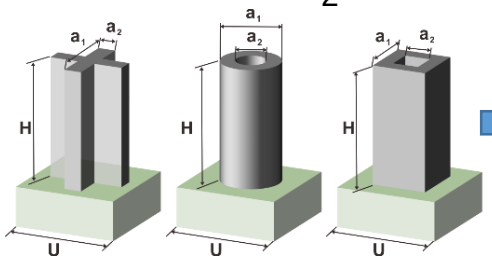
Chen, W. T. et al. Nat. Nanotechnol. 2018.
Shrestha, S. et al. Light-Sci. Appl. 2018.
Wang, S. et al. Nat. Nanotechnol. 2018.

Metalens design with machine learning

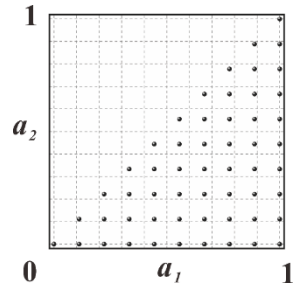
- Artificial neural network

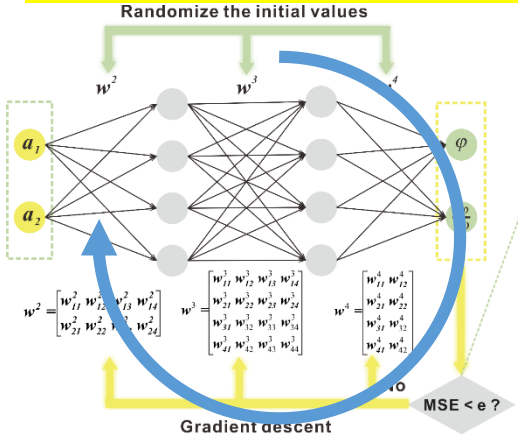
training, validation, test set

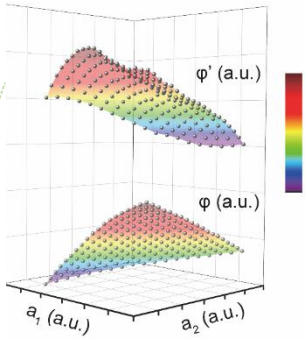
Materials: TiO_2, n=2.4



Patterns



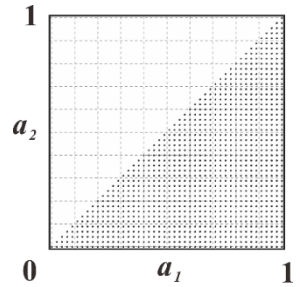




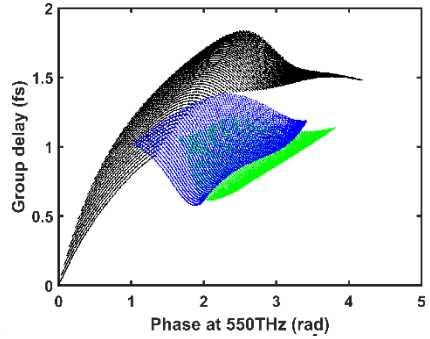
Four-fold symmetry: polarization insensitive

Table S1. Sampling of the three patterns

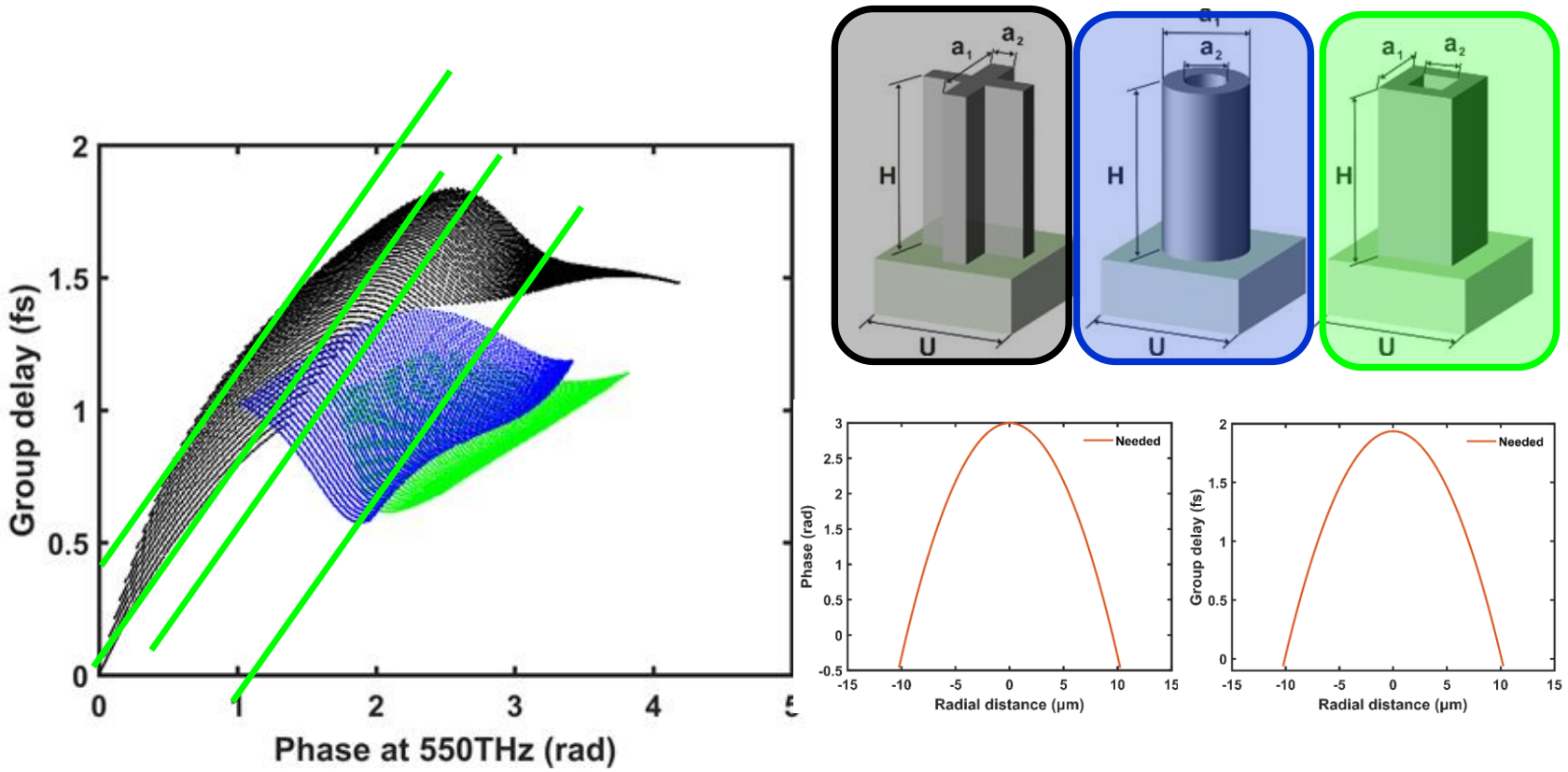
	Shape	Size range (nm)			Sampling step (nm)	Counts
		a1	a2	a1-a2		
Data from ANN	□	180~270	60~150	⩾120	1	4186
	○	180~270	60~150	⩾120		4186
	✚	150~270	50~170	⩾100		7381



Trained neural network



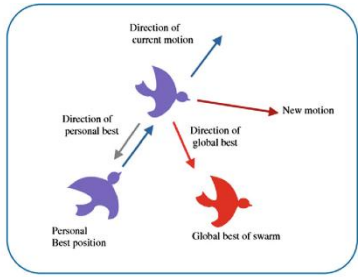
Metalens design with machine learning



Metalens design with machine learning

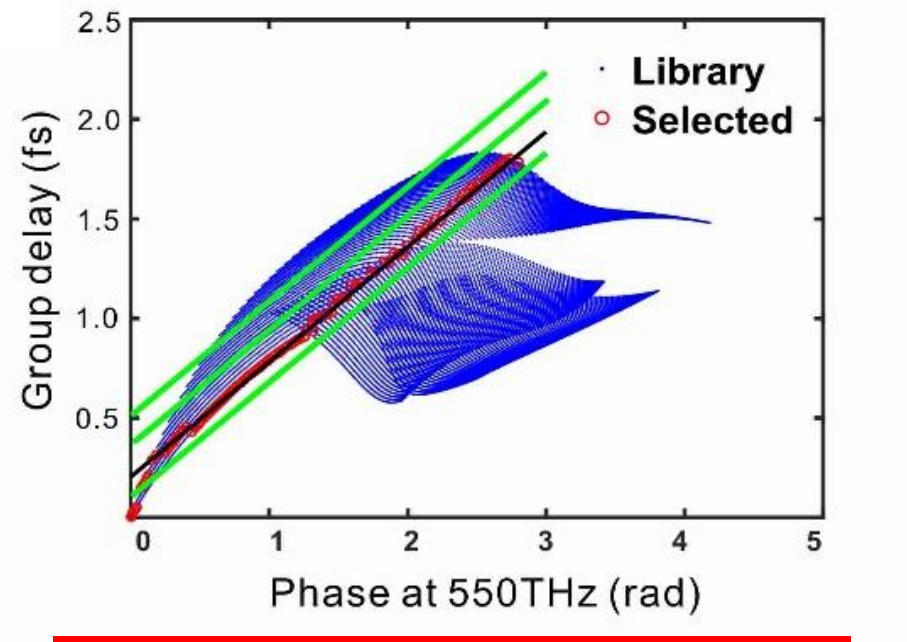
- Selection





Initial database：695 patterns
Simulation time：116h
Training time：20s;
Final database：15753 patterns
Generation time：0.67s
Selection method：particle swarm

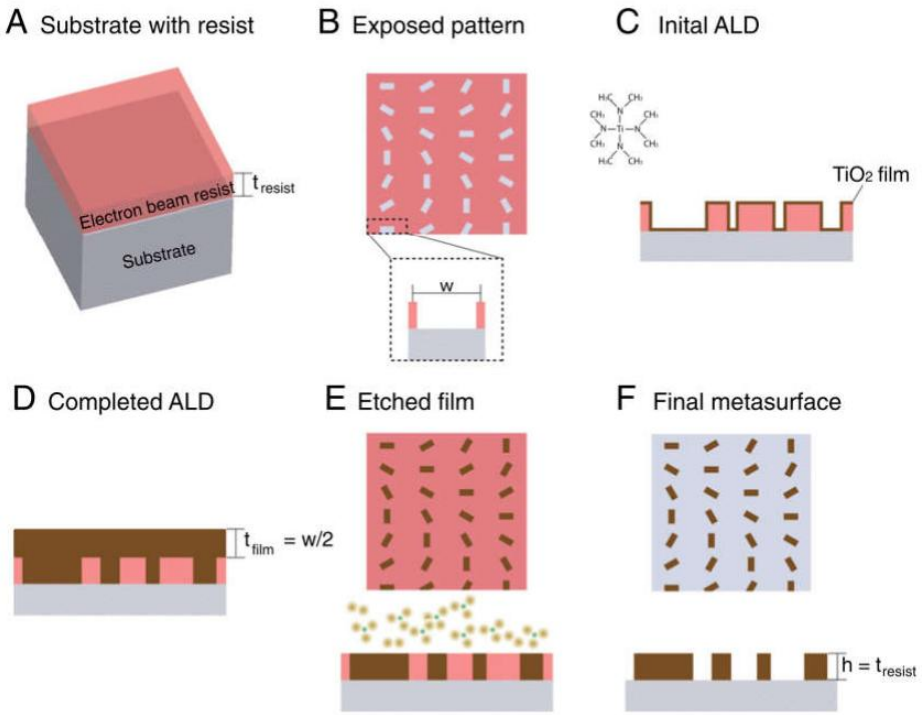
optimization
Searching time：20min

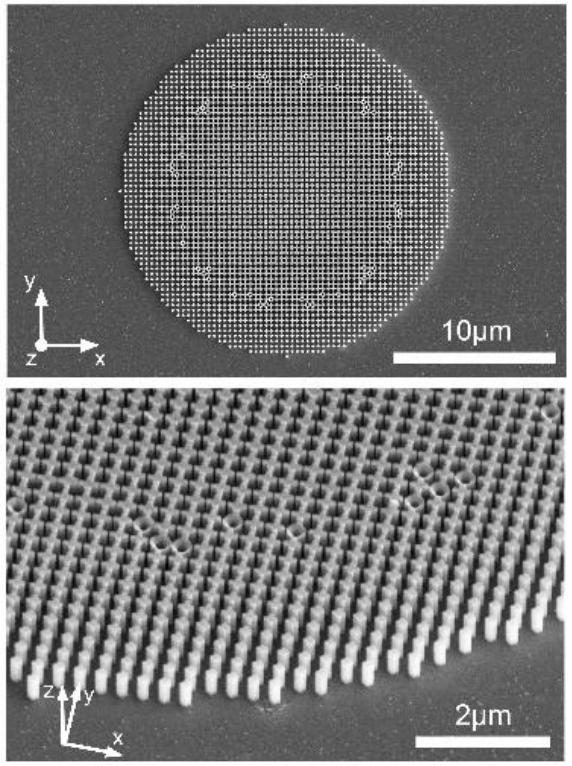


Final database：15753 patterns
Simulation time：2625 h

Metalens Fabrication

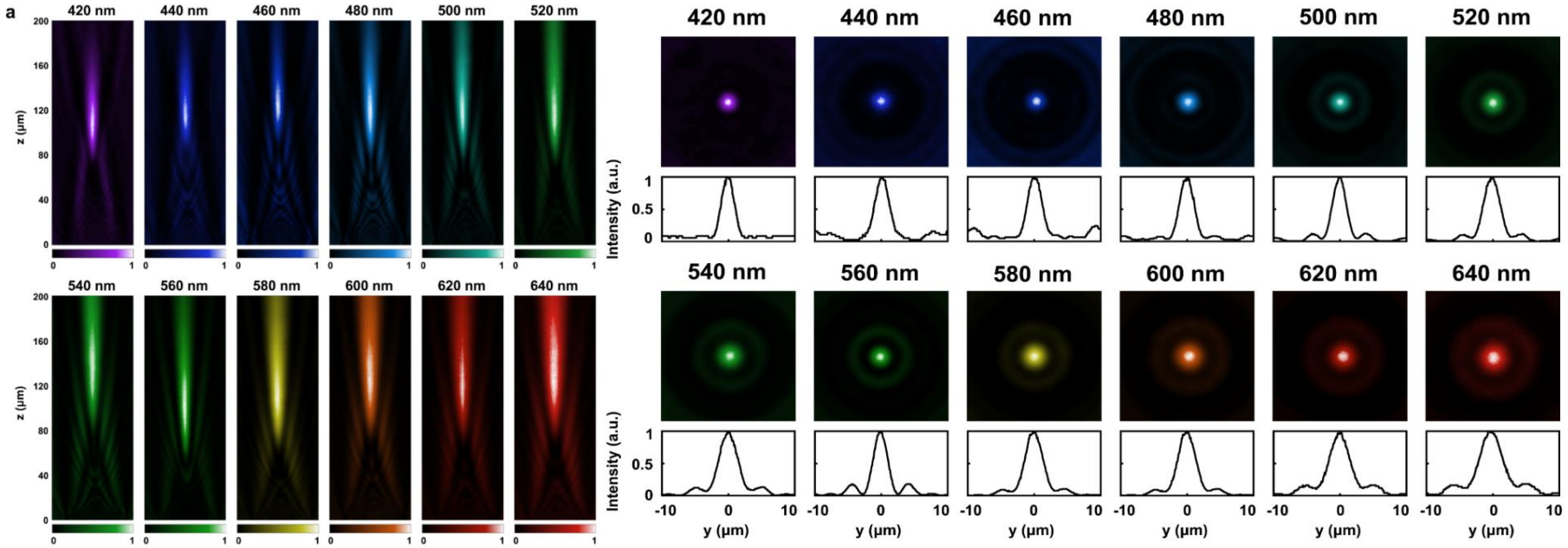
- EBL+ALD





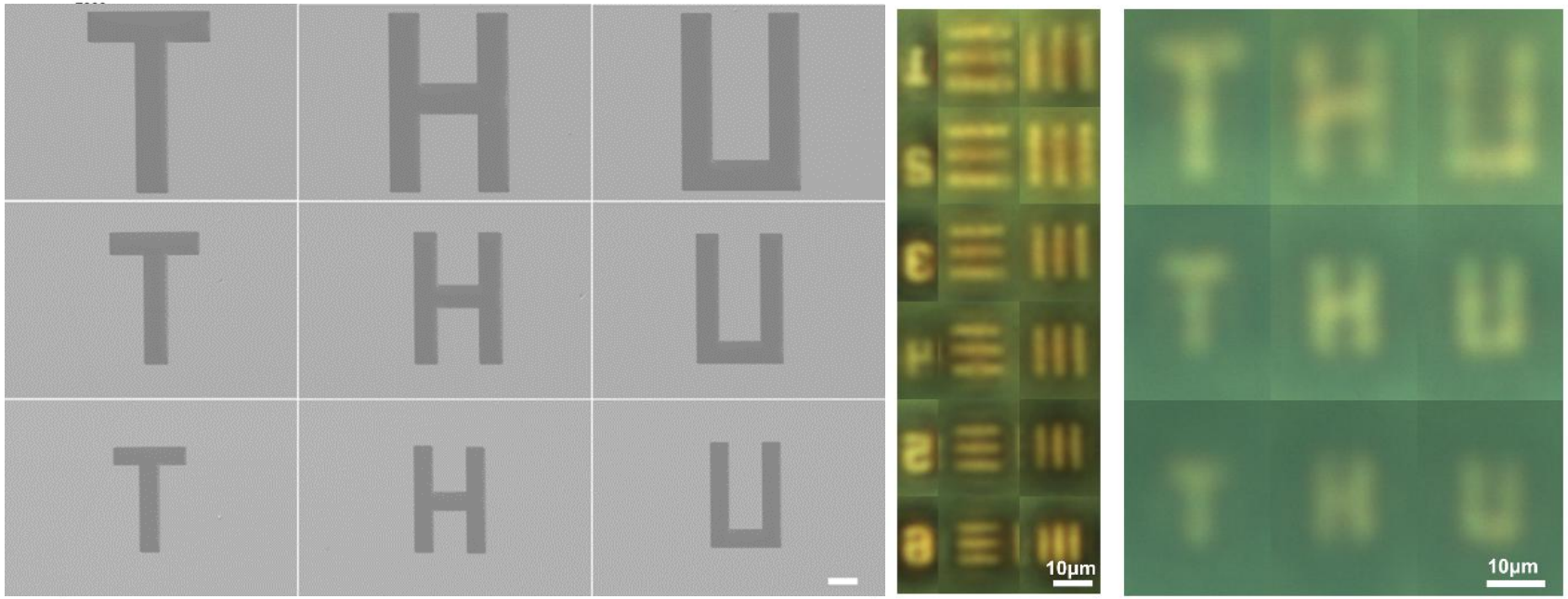
Metalens characterization

- Achromatic focusing



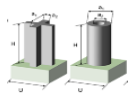
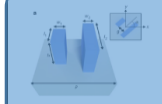
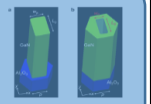
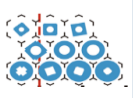
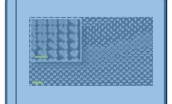
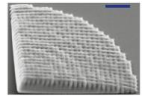
Metalens characterization

- Achromatic Imaging



Metalens characterization

- Performance comparison

Patterns						
Working bandwidth	420-640nm	470-670nm	400-660nm	430-780nm	460-700nm	450–1700 nm
Phase modulation	Propagation phase method	Geometry phase method	Geometry phase method	Propagation phase method	Geometry phase method	Recursive ray-tracing
Thickness	400nm	600nm	800nm	400nm	600nm	1.94μm (non-uniform)
Diameter of a single lens	20.8μm	25.7μm	50μm	14μm	26.4μm	20μm
NA	0.084	0.2	0.106	0.086	0.2	0.27
Efficiency	40.50%	~20%	40%	47%	~30%	~60%
Materials	TiO_2	TiO_2	GaN	SiN	TiO_2	Nanoscribe's IP-DIP photoresist
Polarization dependence	No	Yes	Yes	No	No	No

Much faster design process

Polarization insensitive

Much simpler patterns bring a good performance

Large data library can supply more possibility to build a better lens